

PROVEIT RESEARCH COMPANION

Specialization-Safe Radical Solvers

Exact Bad Characteristics for Subset-Sum Resolvents,
a Fourier–Kummer Atlas, and a Global Obstruction

Research prepared for Vladimir Reshetnikov

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ProveIt research-report collection, continuing the formal project `Algebra/PolynomialFormulas`. Batch 36, manuscript 04 (archive `ProveIt_Radical_Solvers_Research`); pinned to ProveIt `e21766d04`. AI-assisted, unrefereed, not formalized. Notes marked `[write]` were added when the manuscript was written into the collection; see Section 1.5.

Part II (added 29 September 2026), *Optimal Kummer atlases for prime cyclic configurations*: batch 52, manuscript 09 (archive `ProveIt_Optimal_Kummer_Atlases`); pinned to ProveIt `1ee53d57d`; see Section 16.1.

Abstract

A separable polynomial can have an inseparable auxiliary resolvent; a correct radical identity can become unusable on a nonempty exceptional locus; and separately chosen radical branches need not reconstruct one compatible root tuple. We study these three problems in the prime-degree setting of the ProveIt polynomial-formula project.

For equal-cardinality subset sums of the roots of an irreducible prime-degree polynomial, we give an exact, finite cyclotomic-resultant criterion for the characteristics in which collisions are possible. It yields the complete exceptional sets $\{5\}$ for quintic pair sums, $\{2, 7\}$ for septic pair sums, and $\{2, 7, 29\}$ for septic triple sums. In particular, over $\mathbb{F}_{29}(t)$ the separable irreducible polynomial $X^7 - t$ has triple-sum resolvent

$$(Y^7 - 12t)^2(Y^7 - t)(Y^7 - 17t)(Y^7 - 28t).$$

In characteristic 29, collisions occur precisely for translated pure septics; in characteristic two they characterize quotients of additive degree-eight polynomials, up to translation. The explicit example supplies a concrete obstruction to an unrestricted distinct-factor reading of a published septic factor-pattern assertion. A polynomial separating-invariant construction repairs the specialization problem over any sufficiently large field; for septic triples, 1191 distinct parameter values always suffice.

We then prove a branch-coherent Fourier–Kummer reconstruction using one p th root on each nonvanishing Fourier chart. Every nonempty Fourier support occurs for an irreducible rational polynomial with full affine Galois group. Finally, the cyclic quotient of the ordered distinct-root configuration space has Picard group $\mathbb{Z}/p\mathbb{Z}$, and its universal cyclic cover has no global regular Kummer generator. Thus local formulas are not merely an artifact of a poor choice of coordinates. Complete arguments, finite integer certificates, reproducible exact checks, a repository integration plan, and further research questions are included.

Status. This is a mathematical research article with exact computational certificates, not a newly kernel-checked Lean development. The basic Fourier, Galois, resultant, and descent tools are classical. The specific classifications, examples, and synthesis below are proved here; priority over all existing literature has not been established. No claim to have solved the general problem of finding new radical formulas in every degree is made.

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1 Research target, provenance, and contribution boundary

1.1 Why this problem arises in ProveIt

The repository snapshot used for the investigation is commit

e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9.

The root README identifies the audit of Lazard’s solvable-quintic algorithm as an active direction. The polynomial-formula report distinguishes raw invariant equations from root-origin or Fourier evidence, and explicitly records exceptional denominators and the need for coherent radical choices [4, 5, 6]. These are mathematical issues, not only proof-assistant engineering issues.

The inspected source `LazardGeneralFourier.lean` already proves arbitrary-degree character orthogonality, cyclic-shift covariance, and Fourier inversion. It deliberately stops before branch-sensitive recovery of modes from invariants [7]. Accordingly, ordinary discrete Fourier inversion is not presented below as a new result. Our reconstruction uses that existing interface and adds a pivot chart, invariant ratios, and a proof that every choice of the one remaining radical changes only the cyclic ordering of the roots.

[write] **Repository context the manuscript did not cite.** Checked against the tree, which is unchanged in `Algebra/PolynomialFormulas` between the pin and batch 36.

- *The Rocq twin.* Beside the Lean module `Algebra/PolynomialFormulas/Lean/PolynomialFormulas/LazardGeneralFourier.lean` (`character_orthogonality`, l. 49; `dft_shift`, l. 57; `invDFT_dft`, l. 98; the scope boundary quoted above is its docstring, l. 18–21) the project has `Algebra/PolynomialFormulas/Coq/LazardGeneralFourier.v`, proving the same arbitrary-degree core over any MathComp field: `lazard_general_character_orthogonality` (l. 96), `lazard_general_dft_shift` (l. 128) and `lazard_general_invDFT_dft` (l. 207), with the coordinatewise `lazard_general_invDFT_dft_coordinate` (l. 179) [8]. The claim crosswalk (`Algebra/PolynomialFormulas/LazardPaperClaimCrosswalk.md`, l. 199) records both focused modules as kernel-green with their public aggregate/audit pending, and states the same boundary: arbitrary-degree branch-sensitive invariant recovery is deliberately outside the general Fourier modules, while the quintic invariant recovery is covered by a separate five-coordinate chain.
- *Lazard’s own one-radical scheme.* The pivot chart and invariant ratios are not new in the quintic case. The project report `Algebra/PolynomialFormulas/LazardQuinticFormalization.tex` (l. 606–612) records Lazard’s invariants $S_1 = s_1^5$, $S_2 = s_2 s_1^3$, $S_3 = s_3 s_1^2$, $S_4 = s_4 s_1$ and his reconstruction “once $P_1^5 = S_1$ is chosen with $P_1 \neq 0$ ”, with $P_2 = S_2/P_1^3$, $P_3 = S_3/P_1^2$, $P_4 = S_4/P_1$ [5]. This is the case $p = 5$, pivot $j = 1$ of Theorem 8.1: with $\zeta = \omega^{-1}$ one has $u_k = s_k$, $a_1 = S_1$, $c_{1,k} a_1 = S_k$ and $\rho = P_1$ (dictionary in Section 1.4). What the chart theorem adds is the arbitrary odd prime p , the arbitrary pivot j , the statement that *every* branch of the one radical yields the cyclic shift x_{i+rj-1} , the transition rules on overlaps, and Corollary 8.2. A name search of the Lean quintic modules at the write (for `shift` and `rotate`) found no theorem that every fifth-root choice gives a cyclic shift; that search is not a proof of absence.

Three precise research problems guide the article. First, determine exactly when a standard subset-sum resolvent can fail to separate subsets despite separability and irreducibility of the original polynomial. Second, replace an unsafe invariant by one with a proved specialization guarantee. Third, understand whether one can eliminate exceptional radical charts globally, or whether there is a structural obstruction. These are the questions formulated and answered here; they are not attributed to an existing issue or to a named open-problem list.

1.2 Relation to the published formulas

Lazard’s 2004 chapter gives explicit formulas for solvable quintics [1]. Faggal and Lazard’s 2014 article also treats septics, using sums of three roots and their factorization patterns; its specialized septic construction assumes characteristic different from 2, 3, 7 [2]. Thus the existence of a radical treatment of solvable septics is not a new claim of this article.

Our characteristic-29 example concerns the distinction between an orbit product and a squarefree resolvent with distinct conjugate values. It does not, on its own, disprove an abstract radical-solvability theorem. In particular, we do not infer that every later identity in that article fails. Section 6 states the precise implication and its limits.

1.3 What is established, and by what kind of evidence

Result	Mathematical content	Evidence here
Exact bad-characteristic criterion	Necessary and sufficient cyclotomic norm condition, with irreducible witnesses	General proof
Small-degree classification	Complete exceptional sets for $(p, k) = (5, 2), (7, 2), (7, 3)$	Proof plus finite integer tables
Characteristic-29 witness	Explicit repeated-factor triple-sum resolvent	Direct field calculation
Septic collision rigidity	Complete root-configuration classification in characteristics 2 and 29	General proof and finite norm certificate
Separating repair	Characteristic-independent subset descriptors and a finite parameter bound	General proof
One-radical charts	Coherent reconstruction and transition functions	General proof
Support realization	Every nonempty support with full affine Galois group	Explicit rational construction
Global obstruction	$\text{Pic}(A^{C_p}) \cong \mathbb{Z}/p\mathbb{Z}$ and no eigenunit	Algebraic descent and unit calculation
Lean integration	Proposed declarations and dependencies	Plan only; no new Lean build

The universal theorems do not depend on numerical experiments. The three small-degree classifications use a fully explicit finite calculation: all 850 unordered pairs are checked by exact resultants, and the 29 affine-orbit representatives are checked independently by integer determinants. The provided standard-library auditor checks the finite certificates without SymPy. This is still an ordinary computational proof boundary, not a proof-assistant kernel boundary.

1.4 [write] Conventions and notation

No symbol of the delivered manuscript was renamed and no normalization was changed. Several letters carry more than one meaning, inside the article or between the article and the formal project it continues; each is unambiguous in context, and the table fixes the readings.

Symbol	Meaning here	Not to be confused with
u_j, ζ	Fourier mode $u_j = \sum_i \zeta^{-ij} x_i$ (negative exponent), (14)	Lazard's $s_k = \sum_j \omega^{kj} x_j$ (positive exponent), in the project report and in <code>LazardGeneralFourier.lean</code> , whose character is called ψ . <i>True</i> : with $\zeta = \omega^{-1}$, $u_k = s_k$. <i>False reading</i> : $u_k = s_k$ for $\zeta = \omega$; then $u_k = s_{p-k}$.
$a_j, c_{j,k}, \rho$	Pivot power u_j^p , invariant ratios $u_k/u_j^{e_j(k)}$, a root of $T^p - a_j$ (Theorem 8.1)	For $p = 5, j = 1$: $a_1 = S_1$, $c_{1,k}a_1 = S_k$, $\rho = P_1$ in Lazard's notation [5]; $c_{1,k}$ itself is <i>not</i> S_k .
$e_j(k)$	The integer in $\{1, \dots, p-1\}$ with $je_j(k) \equiv k$ (Section 8)	e_r , the r th elementary symmetric function (Sections 5, 7).
$\mathcal{B}_{p,k}$	Set of bad primes (Section 3)	$B_{n,k} = (k-1)\binom{N}{2}$, an integer bound (11).
A, B	Subsets of indices (Sections 2–7)	B : the fixed field $E^{\langle \sigma \rangle}$ (Section 8), B_0, B in Section 9; $A, B = A^{C_p}$: rings in Section 12.
E	A field (Sections 8, 9); $E(A)$: the descriptor vector (Section 7)	Lazard's invariant denominator E in the project report.
k	Subset size (Sections 2–7); mode index in $c_{j,k}$; algebraically closed field (Section 12)	—
N	Norm $N_{A,B}$ (3); $N = \binom{n}{k}$ (Section 7); a minimal normal subgroup (Lemma 9.1)	—
f, g, m	The polynomial; but $f = e_h(k)$, $g = e_j(k)$ and $m = (ef - g)/p$ in (19)	m_{ij} , entries of the multiplication matrix M_q .
T	Polynomial variable (Sections 3, 4), Kummer variable in $T^p - a_j$	Lazard's quadratic-stage quantity T in the project report.
ψ	The $\mathbb{F}_2$ -linear map in the proof of Theorem 11.2	The additive character ψ of <code>LazardGeneralFourier.lean</code> .
L	Splitting field; L_W additive root polynomial; L_r character eigenspace modules (Section 12)	—
Δ	Discriminant $\prod_{i < j} (x_i - x_j)^2$ (25)	—
c	$c_r(A)$ (Theorem 5.1); $c_{j,k}$; c_k (Theorem 10.1); coefficient vector (Section 3); a constant in (27)	—

1.5 [write] Provenance of this report

This report is built from **one** manuscript: batch 36, manuscript 04 of ProveIt's incoming-reports intake, delivered as the archive `ProveIt_Radical_Solvers_Research` in commit `e13affd32`, pinned to ProveIt commit `e21766d04c2b8a9b2cd43e563024b3bd1b9`, and placed in commit `1a1396d4d`. It is filed in ProveIt's research-report collection and continues the formal project `Algebra/PolynomialFormulas`; it shares no theorem with any other report. Every statement, proof, table, example, remark, question and limitation of the delivered text is printed. No merge choice arose. (This paragraph describes Part I. Part II, added on 29 September 2026, is built from a second manuscript, batch 52 manuscript 09; its provenance is Section 16.1.)

The write changed only the following. Every label carries the prefix `rss:`. The title-page provenance line, Sections 1.4, 1.5 and 1.6, the unnumbered [write] notes, and the bibliography entries [8], [9], [10] and [11] were added. The notes use no theorem counter, so every theorem, equation and table number is that of the delivered 29-page PDF. The delivered text names the delivery layout: its files `certificates/X` are shipped as `data/X`, its two scripts as `code/verify_results.py` and `code/audit_certificates.py`, and its `requirements.txt` as `data/requirements.txt`; the delivered PDF and README are not shipped. Appendix A gives commands that work with the shipped names.

1.6 [write] Formal status and what is not claimed

- **Not formalized.** No statement of this report is formalized in Lean or Rocq, and no `rss` label has a Lean or Rocq counterpart. That the report continues the `Algebra/PolynomialFormulas` development confers no formal status on it. The project's own nearby declarations are named where they are used: the arbitrary-degree Fourier core in Lean and Rocq (Section 1), sextic descriptor separation (Section 7), the cyclic Kummer generator (Section 8) and two positive-characteristic counterexamples (Section 3). None of them states a theorem of this report.
- **Integration plan.** Section 13 is a plan. None of it is formalized, and no Lean or Rocq code is delivered or shipped; its declaration names are proposals.
- **Evidence.** The general theorems rest on written proofs. The finite tables rest on exact computations by an ordinary Python program and an independent standard-library auditor, not a proof-assistant kernel. The Fourier checks cover coefficient-one supports only. Nothing computational checks the Picard-group and eigenunit section (Section 12), the transition rule (19), the Galois groups $\text{AGL}_1(\mathbb{F}_p)$, or the rigidity theorems beyond their finite character-pattern counts.
- **The source's own non-claims** stand as printed: the Status paragraph after the abstract (not a kernel-checked development; classical tools are not new; priority over the literature not established; no general radical formulas in every degree); the limits of Section 6 (distinct-factor separation only, not a complete audit of the published algorithm); the assumed invariant data of Theorem 8.1 (Section 8); relative minimality only in Corollary 8.2; minimality of the coordinate-pivot cover only in Corollary 10.2; what Theorem 12.1 does not rule out; and the status of the research questions (Section 14).
- **Review.** The report is AI-assisted and unrefereed. At the intake the proofs were read, the headline computations were repeated by a separate script, and both delivered suites were rerun on a copy; this is not an independent proof review. Parts of Section 3 (the relation-module viewpoint and characteristic-zero separation) may be close to the classical literature on linear relations among conjugates; this was not checked, and no priority is claimed.
- **Part II** (added 29 September 2026) answers Research question 14.5. It is likewise AI-assisted, unrefereed and not formalized; its own status and non-claims are Section 16.3.

2 Subset-sum resolvents and the correct notion of separation

Let p be an odd prime and let $f \in K[X]$ be monic, separable, and irreducible of degree p . In its splitting field choose a numbering $x_0, \dots, x_{p-1}$ of its roots. For $1 \leq k \leq p-1$ put

$$\mathcal{S}_{p,k} = \{A \subseteq \{0, \dots, p-1\} : |A| = k\}, \quad s_A = \sum_{i \in A} x_i,$$

and define the *orbit product*

$$R_{p,k,f}(Y) = \prod_{A \in \mathcal{S}_{p,k}} (Y - s_A). \quad (1)$$

Galois automorphisms permute the subsets, so this monic polynomial belongs to $K[Y]$. Its degree, with multiplicity, is always $\binom{p}{k}$.

Definition 2.1. The k -subset sums of f *separate* if $s_A \neq s_B$ for every two distinct members of $\mathcal{S}_{p,k}$. Equivalently, $R_{p,k,f}$ is separable, or $\gcd(R_{p,k,f}, R'_{p,k,f}) = 1$.

This is stronger than separability of f . The latter distinguishes the individual x_i , not all sums made from them. It is also important not to call (1) the minimal polynomial of a single value without qualification. If the Galois group has several orbits on subsets, then even a separating orbit product is reducible. If distinct subsets have the same value, the product has repeated factors.

Lemma 2.2 (Orbit-to-factor principle). *Let a finite Galois group G act on a finite set $\mathcal{S}$, and let $A \mapsto v_A$ be a G -equivariant map into the Galois extension. If this map is injective, the irreducible factors over the fixed field of $\prod_{A \in \mathcal{S}} (Y - v_A)$ are indexed by the G -orbits on $\mathcal{S}$, and their degrees equal the orbit sizes.*

Proof. The conjugates of v_A are exactly v_{gA} for $g \in G$. Injectivity identifies its stabilizer with that of A . Its minimal polynomial is therefore the product over the orbit of A , with distinct roots. Different orbits have disjoint value sets, and multiplication gives the assertion. $\square$

Without injectivity, two different subset orbits can produce the same irreducible factor. This is exactly the mechanism in the characteristic-29 example, not a loss of separability of the original degree-seven equation.

Complementation gives a useful symmetry:

$$s_{A^c} = \text{Tr}(x_0) - s_A, \quad \text{Tr}(x_0) = \sum_{i=0}^{p-1} x_i.$$

Consequently separation for k is equivalent to separation for $p - k$. The cases $k = 1, p - 1$ always separate. The interesting cases are $2 \leq k \leq p - 2$.

3 An exact cyclotomic criterion for bad characteristics

For distinct $A, B \in \mathcal{S}_{p,k}$ define

$$q_{A,B}(T) = \sum_{i \in A} T^i - \sum_{i \in B} T^i, \quad \Phi_p(T) = 1 + T + \cdots + T^{p-1}, \quad (2)$$

and the positive integer

$$N_{A,B} = |\text{Res}_T(\Phi_p, q_{A,B})|. \quad (3)$$

Equivalently, $N_{A,B}$ is the absolute norm of $q_{A,B}(\zeta)$ from $\mathbb{Q}(\zeta)$, for a primitive p th root of unity ζ .

Lemma 3.1. *Every integer $N_{A,B}$ in (3) is nonzero. It is divisible by p .*

Proof. The polynomial $q = q_{A,B}$ is nonzero, has degree at most $p - 1$, and $q(1) = 0$. If q were divisible by Φ_p over $\mathbb{Q}$, it would equal $c\Phi_p$ for some constant c , and $0 = q(1) = cp$ would force $c = 0$. Irreducibility of Φ_p over $\mathbb{Q}$ therefore gives a nonzero resultant. Modulo p , $\Phi_p = (T - 1)^{p-1}$ and $q(1) = 0$, so the reductions have a common root. Their resultant is zero modulo p . $\square$

Definition 3.2. For $2 \leq k \leq p - 2$, let $\mathcal{B}_{p,k}$ be the set of prime divisors of the integers $N_{A,B}$, as $\{A, B\}$ runs over all unordered pairs of distinct k -subsets. It is a finite, effectively computable set.

Theorem 3.3 (Exact universal separation criterion). *Let p be an odd prime and $2 \leq k \leq p - 2$.*

- (i) *In characteristic zero, the k -subset sums of every separable irreducible degree- p polynomial separate.*
- (ii) *In characteristic $\ell \neq p$, the following are equivalent:*

- (a) for every field K of characteristic ℓ and every separable irreducible degree- p polynomial over K , the k -subset sums separate;
- (b) $\ell \notin \mathcal{B}_{p,k}$.

(iii) In characteristic p , universal separation fails.

Thus $\mathcal{B}_{p,k}$ is exactly the set of characteristics in which a counterexample can exist.

3.1 Proof of the safe direction

Let L/K be the splitting field of f , and $G = \text{Gal}(L/K)$. Irreducibility makes the root action transitive, so p divides $|G|$. Cauchy's theorem supplies an element of order p . A nonidentity order- p permutation on p letters is a p -cycle. We may consequently number the roots so that

$$\sigma(x_i) = x_{i+1}, \quad i \pmod{p}.$$

All pairs of subsets occur in the definition of $\mathcal{B}_{p,k}$, so changing the numbering does not change the criterion.

Suppose that $s_A = s_B$, and write $q = q_{A,B}$. Then

$$q(\sigma)x_0 = 0.$$

In characteristic zero, Lemma 3.1 gives $\gcd(q, \Phi_p) = 1$ in $\mathbb{Q}[T]$. In characteristic $\ell \notin \mathcal{B}_{p,k}$, the resultant is nonzero modulo ℓ , so the same coprimality holds in $\mathbb{F}_\ell[T]$. In either case, over the prime field there are polynomials a, b with

$$a(T)q(T) + b(T)\Phi_p(T) = 1.$$

Apply the corresponding operator identity to $(\sigma - 1)x_0$. Both terms on the left vanish: the first because $q(\sigma)x_0 = 0$, and the second because

$$\Phi_p(\sigma)(\sigma - 1) = \sigma^p - 1 = 0.$$

Hence $(\sigma - 1)x_0 = 0$, contradicting $x_1 \neq x_0$. This proves (i) and the implication (b) $\Rightarrow$ (a) of (ii).

Notice that no irreducibility assertion about Φ_p over K was used. In particular, the safe direction remains valid when K already contains all p th roots of unity.

3.2 Proof that every bad prime really occurs

Suppose $\ell \neq p$ divides $N_{A,B}$. The reductions of Φ_p and $q_{A,B}$ have a common root ζ in $\overline{\mathbb{F}_\ell}$. Since $\ell \neq p$, this is a primitive p th root of unity. Put

$$K = \mathbb{F}_\ell(\zeta)(t), \quad f(X) = X^p - t,$$

where t is transcendental. The polynomial is Eisenstein at t in $\mathbb{F}_\ell(\zeta)[t][X]$, hence irreducible over K . Its derivative is pX^{p-1} , so it is separable. If $\alpha^p = t$, its roots are $x_i = \alpha\zeta^i$, and

$$s_A - s_B = \alpha q_{A,B}(\zeta) = 0.$$

Moreover, $K(\alpha)/K$ is already its splitting field and has cyclic Galois group of order p . Thus the unsafe characteristic is witnessed even by a very simple solvable polynomial. This proves the converse in (ii).

For (iii), use the Artin–Schreier polynomial

$$f(X) = X^p - X - t \in \mathbb{F}_p(t)[X].$$

It is separable. To see its irreducibility, first note that $t \neq b^p - b$ for any $b \in \mathbb{F}_p(t)$: a pole of $b^p - b$ has order divisible by p , whereas t has a pole of order one at infinity. If α is a root, all its roots $\alpha + i$ lie in $\mathbb{F}_p(t)(\alpha)$; the Galois group embeds into the additive group $\mathbb{F}_p$ and therefore has order one or p . The absence of a root in the base field rules out order one.

For a k -subset A , its sum is

$$k\alpha + \sum_{i \in A} i.$$

There are at most p such values, but $\binom{p}{k} > p$ in the stated range of k . Some distinct subsets have equal sums. This finishes the proof of Theorem 3.3. $\square$

Remark 3.4 (The exact logical scope). A prime in $\mathcal{B}_{p,k}$ does not mean that every polynomial in that characteristic has a collision. It means that a collision is possible and that a characteristic-only universal guarantee is false. A particular resolvent can still be tested for separability by a gcd computation.

[write] **Related positive-characteristic material in the project.** Two formal developments of `Algebra/PolynomialFormulas` concern neighbouring phenomena at the prime 3 [11]; neither states a theorem of this section.

- **LazardAlternatingResolventCounterexample** (Lean and Rocq; claim crosswalk l. 202) refutes the assertion that the alternating resolvent $X^2 - \Delta$ of an irreducible polynomial is always separable: over $\mathbb{F}_3(t)$ the cubic $X^3 - t$ is irreducible but *inseparable* (derivative zero), its discriminant vanishes, and its alternating resolvent is X^2 . There the source polynomial itself is inseparable. The characteristic-29 phenomenon of Theorem 5.1 is different: the septic $X^7 - t$ is separable, and only the auxiliary resolvent acquires repeated factors.
- **LazardArtinSchreierRadicalCounterexample** (Lean and Rocq; crosswalk l. 200) uses the Artin–Schreier cubic $X^3 - X - t$, the case $p = 3$ of the witness $X^p - X - t$ in part (iii) above, to show that a cyclic, hence solvable, Galois group in characteristic 3 need not give solvability by radicals. It concerns radical towers, not subset-sum collisions.

3.3 A general relation-space interpretation

The same proof applies to a prescribed coefficient vector $c = (c_0, \dots, c_{p-1}) \in \mathbb{Z}^p$ with $\sum c_i = 0$ and $q_c(T) = \sum c_i T^i \neq 0$. Outside the prime divisors of $\text{Res}(\Phi_p, q_c)$, the relation $\sum c_i x_i = 0$ is impossible for a separable irreducible degree- p polynomial. Conversely, every non- p prime divisor gives a pure-binomial witness for that relation after adjoining the necessary roots of unity. Thus the subset-sum theorem is one instance of an exact integral relation-obstruction method, not merely a collection of examples.

For a fixed characteristic different from p , relations among the cyclically ordered roots form a module over $\mathbb{F}_\ell[T]/(T^p - 1)$. The cyclotomic component $\mathbb{F}_\ell[T]/(\Phi_p)$ detects the nonconstant part. Our resultant test asks whether a prescribed small coefficient vector can vanish on this component. This viewpoint suggests extensions to cyclic codes and more general weighted resolvents, discussed in Section 14.

4 Complete classifications in degrees five and seven

Theorem 4.1 (Small-degree exceptional sets). *The exact exceptional characteristics are*

$$\boxed{\mathcal{B}_{5,2} = \{5\}}, \quad \boxed{\mathcal{B}_{7,2} = \{2, 7\}}, \quad \boxed{\mathcal{B}_{7,3} = \{2, 7, 29\}}.$$

The complementary subset sizes have the same exceptional sets.

We give the finite certificate behind this theorem explicitly. It is useful both for independent review and for a future formalization.

4.1 Reduction by the affine group

The affine group $\text{AGL}_1(\mathbb{F}_p)$ acts on subset labels by $i \mapsto ai + b$, where $a \neq 0$. For a primitive p th root ζ ,

$$q_{aA+b, aB+b}(\zeta) = \zeta^b q_{A,B}(\zeta^a).$$

The cyclotomic field automorphism $\zeta \mapsto \zeta^a$ preserves absolute norms, and ζ^b has norm one. Interchanging A and B changes only a sign. Therefore $N_{A,B}$ is constant on affine orbits of unordered pairs.

For each pair, choose the lexicographically smallest representative of its affine orbit. In the following tables, 012 denotes $\{0, 1, 2\}$, and similarly for other strings. The orbit sizes add to the number of all unordered pairs; disjointness is checked by the displayed normalization rule. Each norm can be computed as an integer determinant without algebraic number arithmetic.

4.2 The determinant certificate

Write

$$q(T)T^j \equiv \sum_{i=0}^{p-2} m_{ij}T^i \pmod{\Phi_p(T)}, \quad 0 \leq j \leq p-2.$$

Multiplication by q in $\mathbb{Z}[T]/(\Phi_p)$ has matrix $M_q = (m_{ij})$. Its determinant is $\text{Res}(\Phi_p, q)$. Reduction uses only

$$T^p = 1, \quad T^{p-1} = -(1 + T + \cdots + T^{p-2}).$$

These identities supply a particularly small exact certificate.

For the exceptional septic pair $A = 012$, $B = 346$, the matrix is

$$M_q = \begin{pmatrix} 2 & -1 & 1 & 0 & -2 & 0 \\ 2 & 1 & 0 & 1 & -2 & -2 \\ 2 & 1 & 2 & 0 & -1 & -2 \\ 0 & 1 & 2 & 2 & -2 & -1 \\ 0 & -1 & 2 & 2 & 0 & -2 \\ 1 & -1 & 0 & 2 & 0 & 0 \end{pmatrix}, \quad |\det M_q| = 203 = 7 \cdot 29. \quad (4)$$

The integer matrices for every other row are included in `certificates/cyclotomic_norm_certificates.json`.¹

A	B	Orbit size	$N_{A,B}$
01	02	20	5
01	04	10	5
01	23	10	5
01	24	5	25
Total, quintic pairs		45	

A	B	Orbit size	$N_{A,B}$
01	02	42	7
01	03	42	7
01	06	21	7
01	23	21	7
01	24	42	56
01	26	21	49
01	34	21	7
Total, septic pairs		210	

¹[write] Shipped in this report as `data/cyclotomic_norm_certificates.json`.

Table 1: All affine orbits of unordered pairs of septic triples.

Row	A	B	Orbit size	$N_{A,B}$
1	012	013	42	7
2	012	014	42	7
3	012	015	42	7
4	012	016	21	7
5	012	023	42	7
6	012	034	42	7
7	012	035	42	56
8	012	045	42	7
9	012	046	42	49
10	012	056	21	7
11	012	134	42	56
12	012	136	21	49
13	012	345	21	7
14	012	346	42	203
15	013	015	21	7
16	013	026	42	56
17	013	046	21	7
18	013	245	7	343
Total			595	

Proof of Theorem 4.1. The tables exhaust the pair sets of sizes $\binom{10}{2} = 45$, $\binom{21}{2} = 210$, and $\binom{35}{2} = 595$. The norms occurring in the three cases are, respectively,

$$\{5, 25\}, \quad \{7, 49, 56\}, \quad \{7, 49, 56, 203, 343\}.$$

Their prime divisors are exactly the displayed exceptional sets. Theorem 3.3 supplies both the universal safe conclusion and the existence of counterexamples in every exceptional characteristic. $\square$

As an additional checksum on the finite enumeration, the complete norm multiplicities are

$$\begin{aligned} (5, 2) : & \quad 5^{[40]}, 25^{[5]}, \\ (7, 2) : & \quad 7^{[147]}, 49^{[21]}, 56^{[42]}, \\ (7, 3) : & \quad 7^{[357]}, 49^{[63]}, 56^{[126]}, 203^{[42]}, 343^{[7]}. \end{aligned}$$

Here $n^{[m]}$ means that norm n occurs m times, not an exponentiation. The independent auditor regenerates the orbits and multiplication matrices, checks every determinant, and verifies that their union is exactly the full pair set. No inference from a sample of pairs is being made.

5 Explicit exceptional-characteristic witnesses

5.1 The characteristic-29 septic

Let $K = \mathbb{F}_{29}(t)$, where t is transcendental, and let $\alpha^7 = t$. Take $\zeta = 23 \in \mathbb{F}_{29}$. Its successive powers are

$$(1, 23, 7, 16, 20, 25, 24), \quad \zeta^7 = 1.$$

They are distinct, so ζ is primitive. The polynomial

$$f(X) = X^7 - t$$

is irreducible by Eisenstein, is separable because its derivative is $7X^6$, and has splitting field $K(\alpha)$ with cyclic group C_7 . Its roots are $x_i = \alpha\zeta^i$.

The following equality is entirely explicit:

$$1 + 23 + 7 \equiv 16 + 20 + 24 \equiv 2 \pmod{29}.$$

Thus

$$x_0 + x_1 + x_2 = x_3 + x_4 + x_6. \quad (5)$$

The roots in the two triples are all different. This is not a collision caused by an overlap or by a repeated root of f .

Theorem 5.1 (Exact repeated-factor resolvent). *Over $\mathbb{F}_{29}(t)$,*

$$\begin{aligned} R_{7,3,X^7-t}(Y) &= (Y^7 - 12t)^2(Y^7 - t)(Y^7 - 17t)(Y^7 - 28t) \\ &= (Y^7 - 12t)(Y^{28} - t^4). \end{aligned} \quad (6)$$

Every displayed degree-seven binomial is irreducible. The orbit product has degree 35, but its squarefree part has degree 28.

Proof. There are five translation orbits on the 35 triples. For a representative A , set

$$c_r(A) = e_r(\zeta^i : i \in A), \quad r = 1, 2, 3.$$

The representative values and their seventh powers are as follows; the second and third coordinates will also be used for the repair below.

A	c_1	c_2	c_3	c_1^7
012	2	17	16	12
013	11	1	20	12
014	15	10	25	17
015	20	14	24	1
024	28	22	24	28

Translation by b multiplies the sum $c_1\alpha$ by ζ^b . The product for one orbit is therefore $Y^7 - c_1^7t$. Multiplying the five orbit products gives the first line of (6). The four constants 1, 12, 17, 28 are the fourth roots of unity in $\mathbb{F}_{29}$, which gives the second line. Each nonzero binomial $Y^7 - ct$ is Eisenstein at t , hence irreducible. The four constants are distinct, so the asserted squarefree degree follows. $\square$

5.2 Characteristic two: the Fano-plane degeneration

Let $K = \mathbb{F}_8(t)$, let ζ generate $\mathbb{F}_8^\times$, and again let $\alpha^7 = t$. The polynomial $X^7 - t$ is separable and irreducible over K . Its seven nonzero root coefficients are all of $\mathbb{F}_8^\times$.

A triple of distinct nonzero field elements has sum zero precisely when it is $\{a, b, a+b\}$. There are $7 \cdot 6/6 = 7$ such triples, the lines of the Fano plane. Multiplication by $\mathbb{F}_8^\times$ permutes the other triples and acts transitively on their nonzero sums. The remaining 28 triples are therefore distributed equally, four per nonzero sum. Hence

$$R_{7,3,X^7-t}(Y) = Y^7(Y^7 - t)^4 \quad \text{over } \mathbb{F}_8(t). \quad (7)$$

The auxiliary product has only eight distinct roots, even though the original polynomial has seven distinct roots and a cyclic Galois group.

For septic pair sums in characteristic two, the same field gives $\zeta^a + \zeta^b = \zeta^c + \zeta^d$ for suitable distinct pairs, in agreement with $2 \in \mathcal{B}_{7,2}$. For instance, in $\mathbb{F}_2[z]/(z^3 + z + 1)$, the pairs of field elements $\{1, z\}$ and $\{z^2, 1 + z + z^2\}$ have equal sum.

5.3 Characteristic seven: an Artin–Schreier degeneration

Over $\mathbb{F}_7(t)$, let $f = X^7 - X - t$ and choose a root α . The irreducibility argument in Section 3 applies, and the roots are $\alpha + i$ for $i \in \mathbb{F}_7$. A triple sum is $3\alpha + c$. Translation of a triple by b changes c by $3b$, so each of the seven values occurs exactly five times. Moreover,

$$(3\alpha)^7 - 3\alpha = 3t.$$

It follows that

$$R_{7,3,X^7-X-t}(Y) = (Y^7 - Y - 3t)^5. \quad (8)$$

Likewise,

$$R_{5,2,X^5-X-t}(Y) = (Y^5 - Y - 2t)^2 \quad \text{over } \mathbb{F}_5(t). \quad (9)$$

These identities make the exceptional prime-degree characteristic concrete, rather than merely noting that Fourier inversion cannot divide by p .

6 The exact implication for published septic factor patterns

Proposition 7 and the subsequent specialized discussion in Faggal–Lazard describe a cyclic septic case using five degree-seven factors; Section 5.2 permits characteristic 29 under its stated exclusions [2]. Equation (6) shows that one cannot universally interpret those factors as five *distinct* irreducible factors attached to five distinguishable subset orbits. Two orbit products coincide.

If factors are counted with multiplicity, our example still has five irreducible factors in that weaker sense. The failure is therefore stated precisely: the unqualified separation needed to identify different subset orbits from different scalar factors is false. Also, the degree-35 orbit product is not a minimal polynomial of any one triple sum; in this example each individual triple sum has degree seven.

The sharp repair for this particular separation step is now known:

$$\text{char } K \notin \{2, 7, 29\}.$$

This assertion follows for *every* irreducible separable septic from Theorem 4.1, without a solvability assumption. When retaining the separate characteristic restrictions of a radical construction, one must take their union with these restrictions. Conversely, an input-specific squarefreeness check is sufficient for the orbit-to-factor principle even in a bad characteristic.

This is not a complete audit of all matching predicates or denominator conditions used in the published algorithm. Additional root relations can have different cyclotomic norm obstructions. Excluding 29 repairs the triple-sum separation claim; it does not by itself certify every subsequent formula. The next section offers a repair that avoids relying on a characteristic-only exclusion at all.

7 A uniform separating-invariant repair

Let $x_1, \dots, x_n$ be pairwise distinct elements of a field extension of K . Fix $1 \leq k < n$, and set $N = \binom{n}{k}$. For each k -subset A let

$$E(A) = (e_1(x_A), \dots, e_k(x_A)),$$

where e_r is the r th elementary symmetric function of the elements whose indices lie in A .

Lemma 7.1 (Subset descriptors). *The map $A \mapsto E(A)$ is injective in every characteristic.*

Proof. Its coordinates determine the monic polynomial

$$\prod_{i \in A} (Z - x_i) = Z^k - e_1(x_A)Z^{k-1} + \dots + (-1)^k e_k(x_A).$$

Equal coordinate vectors give equal polynomials and therefore equal sets of roots. Pairwise distinctness of the original x_i then gives equal index subsets. $\square$

Introduce a parameter u and compress this vector to the scalar

$$h_A(u) = \sum_{r=1}^k u^{r-1} e_r(x_A). \quad (10)$$

For different subsets A, B , the difference $h_A - h_B$ is a nonzero polynomial in u of degree at most $k - 1$.

Theorem 7.2 (Finite separating family). *At most*

$$B_{n,k} = (k-1) \binom{N}{2}, \quad N = \binom{n}{k}, \quad (11)$$

values of u in an algebraic closure of the field containing the x_i can cause two of the $h_A(u)$ to coincide. Consequently, any $B_{n,k} + 1$ distinct values in K include one for which all the $h_A(u)$ are distinct.

If the x_i are the roots of a separable polynomial over K , then

$$H_u(Y) = \prod_{|A|=k} (Y - h_A(u)) \in K[Y] \quad (12)$$

for every $u \in K$. At a separating parameter, its irreducible factors recover the exact Galois orbit sizes on k -subsets.

Proof. Each of the $\binom{N}{2}$ nonzero difference polynomials has at most $k - 1$ roots. Taking their union gives (11), with no restriction on characteristic. The product (12) is invariant under every permutation of the x_i , so its coefficients are symmetric polynomials in the roots and hence belong to K . At a separating parameter, Lemma 2.2 applies. $\square$

[write] Existing formal counterparts. The project already formalizes this strategy for sextic block systems in `Algebra/PolynomialFormulas/Lean/PolynomialFormulas/SexticSeparatingInvariants.lean` [9], whose docstring starts from the same observation that scalar pair and triple invariants can collide after specialization. `rootPolynomial_injective` (l. 115) is the argument of Lemma 7.1 for six roots and arbitrary index subsets, over any commutative domain (no characteristic hypothesis). `exists_nat_eval_injective` (l. 1048) separates a finite injective family of multivariate polynomials by one natural-number evaluation point, via the combinatorial Nullstellensatz, over a *characteristic-zero* domain, and `exists_pairDescriptor_separating_evaluation` and `exists_tripleDescriptor_se`

`parating_evaluation` (l. 1074, 1083) apply it to the sextic block descriptors. The resulting “proved terminating unbounded search” is described in `Algebra/PolynomialFormulas/Decidability.md` (l. 284–290). These are analogues of the declarations `elementarySubsetDescriptor_injective` and `separatingParameter_exists` proposed in Section 13. What Lemma 7.1 and Theorem 7.2 add is the general (n, k) setting, the one-parameter compression $h_A(u)$, and an explicit hitting bound $B_{n,k}$ valid in every characteristic.

[write] **Sibling report, added 29 September 2026 (batches 40–41).** The sibling report `sextic-block-resolvent-separators` in this collection (batch 41, manuscript 03, written without knowledge of this report) bounds the parameter search built on those sextic descriptors. It compresses the pair- and triple-partition descriptors of `SexticSeparatingInvariants.lean` on the curve $(u, v) = (t, 2t^2)$ to a quadratic $h_P(t)$ and a linear $k_T(t)$ (its h_P is not the $h_A(u)$ of (10)), counts the collisions by orbits of root permutations, and proves in writing that one of $1, \dots, 196$ separates both families for every separable sextic in characteristic zero, sharply for arbitrary sets of complex test values. It is thus a sextic-partition counterpart of Theorem 7.2, obtained by the kind of orbit count that Research question 14.3 (Section 14) asks for. It concerns neither septic triples nor positive characteristic, so it answers no question of this report. It is not formalized: the Lean search it bounds is still unbounded.

For septics and triples,

$$N = 35, \quad B_{7,3} = 2 \binom{35}{2} = 1190.$$

Thus 1191 *distinct* parameters always suffice whenever the base field has at least that many elements. Over $\mathbb{Q}$ one may use $0, 1, \dots, 1190$. In characteristic 29, those same integer numerals would repeat and would not constitute a valid hitting set. Over $\mathbb{F}_{29}(t)$ one can instead use $1, t, t^2, \dots, t^{1190}$.

Over a finite field with fewer elements, the theorem does not assert the existence of a separating parameter in that field. Passing to a sufficiently large finite extension, or keeping u transcendental, supplies distinct parameters; the associated change of base field must then be recorded when interpreting Galois factor patterns.

7.1 How to compute the repaired product without knowing the roots

Regard $x_1, \dots, x_n$ first as indeterminates. The product (12) is a symmetric polynomial in them, with coefficients in $\mathbb{Z}[u, Y]$. The fundamental theorem of symmetric polynomials expresses it as a polynomial in the elementary symmetric functions of the full root set. Substitution of the coefficients of the original monic polynomial therefore computes H_u over its coefficient field. This is an algebraic construction; it does not require numerical roots or a pre-existing radical solution.

A deterministic specialization procedure is consequently available: compute H_u for each chosen parameter, test $\gcd(H_u, H'_u) = 1$, and stop at the first success. Theorem 7.2 proves termination within the stated number of distinct candidates. It is a correctness and termination bound, not a claim of good asymptotic complexity. Expanding the universal symmetric polynomial can be expensive.

7.2 Power-sum variants and their characteristic restriction

When $1, \dots, k$ are invertible in K , Newton identities show that the vector

$$\left(\sum_{i \in A} x_i, \sum_{i \in A} x_i^2, \dots, \sum_{i \in A} x_i^k \right)$$

also distinguishes subsets. Then one may use

$$\sum_{i \in A} (x_i + ux_i^2 + \dots + u^{k-1}x_i^k).$$

This has the appealing form of an ordinary subset sum after a common polynomial transformation of the roots. The elementary-symmetric version (10), however, avoids division by $2, \dots, k$ and works in every characteristic. These two versions should not be silently identified in small characteristic.

7.3 Repairing the characteristic-29 example explicitly

For $X^7 - t$ over $\mathbb{F}_{29}(t)$, use $u = 1$. Then

$$h_A(1) = c_1(A)\alpha + c_2(A)\alpha^2 + c_3(A)\alpha^3.$$

The 35 triples have distinct coefficient vectors (c_1, c_2, c_3) by Lemma 7.1, applied to the seven distinct powers of ζ . Since $1, \alpha, \dots, \alpha^6$ are linearly independent over $\mathbb{F}_{29}(t)$, all 35 displayed values are distinct. Thus one nonzero parameter already repairs this particular example.

The five now-distinct irreducible factors have the following compact exact specification. For each of the five rows (c_1, c_2, c_3) in the table in Theorem 5.1, take

$$Q_A(Y) = \text{Res}_Z(Z^7 - t, Y - c_1Z - c_2Z^2 - c_3Z^3). \quad (13)$$

Each Q_A is monic of degree seven, and $H_1 = \prod_A Q_A$. They are distinct and irreducible by Lemma 2.2. This is a coefficient-level description of the repaired factors, not merely a statement that some repair exists.

For completeness, the affine solvable septic groups give the following orbit patterns, now valid for any separating equivariant descriptor:

Group	Factor degrees
C_7	7, 7, 7, 7, 7
$C_7 \rtimes C_2$	7, 7, 7, 14
$C_7 \rtimes C_3$	7, 7, 21
$C_7 \rtimes C_6$	14, 21

These counts follow directly by acting on the 35 triples with $i \mapsto ai + b$, where the allowed multipliers form the indicated subgroup of $\mathbb{F}_7^\times$. The verifier also checks these four finite actions.

8 One-radical Fourier–Kummer reconstruction

We now address a different failure mode: reconstructing roots from modes without incompatible radical branches. Assume throughout this section that $\text{char } E \neq p$, that $\zeta \in E$ is a primitive p th root of unity, and that $\sigma \in \text{Aut}(E)$ has order p and fixes ζ . Let $B = E^{(\sigma)}$ and suppose that

$$\sigma(x_i) = x_{i+1}, \quad i \pmod{p}.$$

We use the Fourier convention

$$u_j = \sum_{i=0}^{p-1} \zeta^{-ij} x_i, \quad x_i = \frac{1}{p} \sum_{j=0}^{p-1} \zeta^{ij} u_j. \quad (14)$$

This differs by an index sign from the positive-exponent forward convention in the inspected Lean module. Substituting its character by its inverse aligns the conventions exactly; no mathematical change is involved.

The finite geometric-sum identity gives

$$\sigma(u_j) = \zeta^j u_j, \quad u_0 \in B.$$

If the x_i are not all equal, at least one u_j with $j \neq 0$ is nonzero. This is the only existence condition needed for a pivot chart.

For $j \in \{1, \dots, p-1\}$ define $e_j(k)$ to be the unique integer in $\{1, \dots, p-1\}$ satisfying

$$je_j(k) \equiv k \pmod{p}, \quad 1 \leq k \leq p-1.$$

Theorem 8.1 (Coherent one-radical chart). *Assume $u_j \neq 0$. Put*

$$a_j = u_j^p, \quad c_{j,k} = \frac{u_k}{u_j^{e_j(k)}} \quad (1 \leq k \leq p-1). \quad (15)$$

Then $a_j \in B^\times$, $c_{j,k} \in B$, and $c_{j,j} = 1$. For any root ρ of $T^p - a_j$ in an algebraic closure of E , define

$$y_i = \frac{1}{p} \left(u_0 + \sum_{k=1}^{p-1} \zeta^{ik} c_{j,k} \rho^{e_j(k)} \right). \quad (16)$$

There exists $r \in \mathbb{F}_p$ such that

$$y_i = x_{i+rj-1} \quad \text{for every } i \in \mathbb{F}_p. \quad (17)$$

In particular, all branches give the same unordered root set, with no additional independent p th-root choices.

Proof. The covariance formula gives

$$\sigma(a_j) = a_j, \quad \sigma(c_{j,k}) = \zeta^{k-je_j(k)} c_{j,k} = c_{j,k}.$$

The nonzero denominator is part of the hypothesis, not a convention about division by zero. Since E already contains u_j and all p th roots of unity, every root of $T^p - a_j$ is $\rho = \zeta^r u_j$ for some r . Substitution in (16) yields

$$c_{j,k} \rho^{e_j(k)} = \zeta^{re_j(k)} u_k = \zeta^{rj^{-1}k} u_k.$$

Fourier inversion now gives (17) simultaneously for all i . Thus coherence is derived, not assumed separately for each mode. $\square$

Corollary 8.2 (Cyclic extension generated by a pivot). *Under the hypotheses of Theorem 8.1, $[B(u_j) : B] = p$. If E/B is the cyclic extension of degree p , then $E = B(u_j)$ and $T^p - a_j$ is the minimal polynomial of u_j over B .*

Proof. The p values $\sigma^r(u_j) = \zeta^{rj} u_j$ are distinct. Thus its minimal polynomial over B has at least p roots, while $T^p - a_j$ gives an upper bound of p on its degree. The remaining assertion follows by the degree formula. $\square$

This is a relative minimality statement: the cyclic degree- p step really is nontrivial, and one p th radical suffices over B . It is *not* a claim that an entire radical expression over the original coefficient field uses the smallest possible number of radicals. The repository's warnings about stronger optimality assertions remain relevant.

[write] An abstract counterpart in Rocq. Algebra/PolynomialFormulas/Coq/LazardCyclicKummerGenerator.v proves `cyclic_kummer_generator` (l. 32): if F/E is cyclic Galois of degree n and E contains a primitive n th root of unity, there is a nonzero $x \in F$ with $x^n \in E$ and $F = E(x)$ [10]. It is the abstract existence form of Corollary 8.2; the corollary adds that any nonzero Fourier mode u_j is such a generator.

8.1 Transition maps on overlaps

Suppose $u_j, u_h \neq 0$, and set $e = e_j(h)$. Then

$$u_h = c_{j,h} u_j^e, \quad a_h = c_{j,h}^p a_j^e.$$

A branch $\rho_j^p = a_j$ determines the compatible branch

$$\rho_h = c_{j,h} \rho_j^e, \quad \rho_h^p = a_h. \quad (18)$$

It gives the same reconstructed ordered tuple when the chart coordinate conventions are matched. More explicitly, if $\rho_j = \zeta^r u_j$, then $\rho_h = \zeta^{re} u_h$, and the shift in the h -chart is $re h^{-1} = rj^{-1}$ modulo p .

For the invariant coefficients, let $f = e_h(k)$ and $g = e_j(k)$. The integer

$$m = \frac{ef - g}{p}$$

is well defined. Direct substitution gives the exact transition rule

$$c_{h,k} = c_{j,k} c_{j,h}^{-f} a_j^{-m}. \quad (19)$$

This exhibits an actual atlas, with compatible local coordinates, rather than unrelated formulas whose radical branches must be guessed anew.

8.2 What the chart theorem does not compute

The input to Theorem 8.1 includes the invariant data $u_0, a_j, c_{j,k}$ over B . Constructing these efficiently from the original polynomial coefficients is a separate invariant-theoretic task. The theorem does not hide that task inside a claim of an executable coefficient-only solver. Its immediate use in ProveIt is as a small correctness theorem between an invariant-recovery stage and an already formalized Fourier reconstruction stage.

9 Solvable prime-degree groups and the coefficient-field tower

We record the precise field-theoretic setting in which the cyclic chart arises from a solvable prime-degree equation. The group-theoretic argument is included to make the dependence explicit.

Lemma 9.1 (Solvable transitive groups of prime degree). *A solvable transitive subgroup $G \leq S_p$ is conjugate to $C_p \rtimes C_d \leq \text{AGL}_1(\mathbb{F}_p)$ for some $d \mid p-1$.*

Proof. A transitive action of prime degree is primitive: every block size divides p . Choose a minimal nontrivial normal subgroup N of G . Solvability implies that N is elementary abelian: its commutator subgroup is characteristic in N and normal in G , so minimality and solvability make it trivial. The primary subgroups of the resulting finite abelian group are characteristic, forcing a single prime; multiplication by that prime has characteristic image and cannot be surjective, so this image is zero. Its orbits form a block system, so N is transitive; otherwise every orbit would be a singleton, contrary to faithfulness and $N \neq 1$.

An abelian transitive permutation group is regular. Indeed, an element fixing one point fixes every point by commuting with a transitive group. Thus $|N| = p$ and $N \cong C_p$. Numbering the points by $\mathbb{F}_p$ identifies N with the translations. Its normalizer in S_p is the affine group: a normalizing permutation is determined by the image of 0 and the nonzero multiplier by which it

conjugates a generator of translations. Hence G lies in that normalizer. Its multiplier image is a subgroup of $\mathbb{F}_p^\times$, therefore cyclic of order $d \mid p-1$. Since all translations are present, subtracting the translation part of each affine element gives the asserted semidirect product. $\square$

Now let $f \in K[X]$ be separable irreducible of degree p in characteristic zero, with solvable Galois group $G = C_p \rtimes C_d$, and let L be its splitting field. Write

$$B_0 = L^{C_p}, \quad E = L(\zeta), \quad B = B_0(\zeta).$$

Then B_0/K is cyclic of degree d , and B/K is an abelian Galois extension, as the compositum of B_0/K with a cyclotomic extension. Moreover,

$$[B : K] \leq d(p-1) \leq (p-1)^2.$$

Because $[B : B_0]$ divides $p-1$, it is relatively prime to $[L : B_0] = p$. Thus $L \cap B = B_0$ and

$$[E : B] = p.$$

The cyclic generator on L/B_0 extends to E fixing B , so all the hypotheses of the Fourier–Kummer chart theorem are available after numbering the roots by this generator.

Corollary 9.2 (Field-theoretic solver decomposition). *In characteristic zero, a solvable irreducible prime-degree equation admits the following decomposition: construct the abelian field B of degree at most $(p-1)^2$ over K , select a nonzero Fourier mode, and perform the single cyclic p th-radical reconstruction of Theorem 8.1. For $p=5$ the displayed bound is 16; for $p=7$ it is 36.*

These are field-degree bounds, not optimal radical-count bounds. The abelian stage itself is solvable by radicals by the classical Galois construction after adjoining suitable roots of unity. A fast implementation must exploit the smaller actual Galois group and its invariants rather than blindly realize the worst-case degree bound.

In positive characteristic, the same cyclic p -chart remains valid when $\text{char } K \neq p$ and the stated cyclic extension and root-of-unity data are supplied. The equivalence between solvable groups and ordinary radical towers needs its own characteristic hypotheses and is not inferred merely from group solvability here.

10 Every nonempty Fourier support really occurs

A formula that works for generic tuples need not survive vanishing Fourier modes. The following construction shows that no nonempty pattern can be excluded solely from rational irreducibility or from maximal solvable Galois group.

Theorem 10.1 (Full support realization). *Let p be an odd prime, let $\emptyset \neq S \subseteq \{1, \dots, p-1\}$, and choose nonzero rational numbers c_k for $k \in S$. Let $\alpha = 2^{1/p}$ and let ζ be a primitive p th root of unity. Put*

$$P(Z) = \sum_{k \in S} c_k Z^k, \quad x_i = P(\alpha \zeta^i), \quad 0 \leq i \leq p-1. \quad (20)$$

The x_i are the roots of an irreducible rational polynomial of degree p . Its splitting field has Galois group $\text{AGL}_1(\mathbb{F}_p)$. Relative to the indicated cyclic numbering, its Fourier modes are

$$u_0 = 0, \quad u_k = \begin{cases} p c_k \alpha^k, & k \in S, \\ 0, & k \notin S. \end{cases} \quad (21)$$

Thus all $2^{p-1} - 1$ nonempty supports occur.

Proof. The polynomial $Z^p - 2$ is Eisenstein at 2, so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = p$. Linear independence of $1, \alpha, \dots, \alpha^{p-1}$ shows that $P(\alpha) \notin \mathbb{Q}$. Since p is prime, the tower formula forces

$$\mathbb{Q}(P(\alpha)) = \mathbb{Q}(\alpha).$$

Therefore $P(\alpha)$ has degree p , and its images under the p embeddings of $\mathbb{Q}(\alpha)$ are exactly the displayed distinct x_i . The normal closures of $\mathbb{Q}(P(\alpha))$ and $\mathbb{Q}(\alpha)$ coincide.

For completeness, $[\mathbb{Q}(\zeta) : \mathbb{Q}] = p - 1$ and $\alpha \notin \mathbb{Q}(\zeta)$: otherwise the degree p would divide $p - 1$. Over $\mathbb{Q}(\zeta)$, adjoining α gives the splitting field of $Z^p - 2$; its Galois group embeds in μ_p , and nontriviality forces degree p . Hence the full normal closure has degree $p(p - 1)$ and its natural affine permutation action is all of $\text{AGL}_1(\mathbb{F}_p)$.

Finally, substitute (20) into (14). The geometric sum $\sum_i \zeta^{i(k-j)}$ equals p when $k = j$ and zero otherwise, giving (21). $\square$

Corollary 10.2 (Sharpness of the coordinate-pivot cover). *For a fixed primitive root of unity and a fixed cyclic numbering, none of the $p - 1$ charts $u_j \neq 0$ can be omitted if the chart family is required to cover every irreducible rational degree- p example with full affine Galois group.*

Proof. Use Theorem 10.1 with $S = \{j\}$. Only the j th chart is available. $\square$

The qualifiers matter. This is a minimality statement for this specific coordinate-pivot cover, not a lower bound for every conceivable nonlinear cover. Relabelling the cyclic generator permutes the nonzero Fourier indices; choosing such a relabelling adaptively is another form of chart selection, not a contradiction.

[write] **Nonlinear covers, added 29 September 2026 (batch 52).** Part II supplies that lower bound for covers of the universal cyclic quotient $\text{Spec } B$ of Section 12: every cover by regular Kummer charts, with arbitrary open sets and arbitrary nonlinear character sections, has at least $p - 1$ members (Proposition 21.2 and Corollary 17.2). It concerns Zariski covers of that universal quotient, not the coverage of individual rational examples stated in the corollary above, so it complements the corollary rather than replacing it.

10.1 Concrete rational examples

Taking every $c_k = 1$, elimination of Z from $Z^p - 2 = 0$ and $Y - P(Z) = 0$ gives the following monic polynomials:

$$\begin{aligned} p = 5, S = \{1, 2\} : & Y^5 - 10Y^2 - 10Y - 6, \\ p = 5, S = \{2, 3\} : & Y^5 - 10Y^3 + 20Y - 12, \\ p = 5, S = \{1, 2, 3, 4\} : & Y^5 - 20Y^3 - 60Y^2 - 70Y - 30, \\ p = 7, S = \{1, 2\} : & Y^7 - 14Y^3 - 28Y^2 - 14Y - 6, \\ p = 7, S = \{2, 3\} : & Y^7 - 14Y^4 - 28Y^2 + 28Y - 12, \\ p = 7, S = \{1, \dots, 6\} : & Y^7 - 42Y^5 - 210Y^4 - 490Y^3 \\ & - 630Y^2 - 434Y - 126. \end{aligned}$$

Their irreducibility and Galois groups follow from Theorem 10.1, not from numerical root matching. The exact resultants are also regenerated by the verifier.

For example, in the first septic case, $u_1 = 7\alpha$ and $u_2 = 7\alpha^2$, while all other nonconstant modes vanish. In the $j = 1$ chart,

$$a_1 = 7^7 \cdot 2, \quad c_{1,1} = 1, \quad c_{1,2} = \frac{1}{7}.$$

The formula becomes

$$y_i = \frac{1}{7} \left(\zeta^i \rho + \frac{\zeta^{2i}}{7} \rho^2 \right), \quad \rho^7 = 7^7 \cdot 2.$$

Choosing $\rho = 7\alpha\zeta^r$ recovers $x_{i+r} = \alpha\zeta^{i+r} + \alpha^2\zeta^{2(i+r)}$ exactly.

11 Rigidity of the exceptional septic configurations

The bad-prime calculation can say more than where counterexamples exist. In characteristics 29 and 2, the small number of surviving cyclotomic characters forces the entire root configuration into a special form.

11.1 Characteristic 29: precisely the translated binomial locus

Theorem 11.1 (Complete characteristic-29 collision classification). *Let K be any field of characteristic 29, and let $f \in K[X]$ be monic, separable, and irreducible of degree seven. The following are equivalent:*

- (i) *Two different triples of roots of f have the same sum.*
- (ii) *For some $\eta, b \in K$ with $b \neq 0$,*

$$f(X) = (X - \eta)^7 - b.$$

Whenever these conditions hold, the Galois group of f is C_7 , and

$$R_{7,3,f}(Y) = ((Y - 3\eta)^7 - 12b)((Y - 3\eta)^{28} - b^4). \quad (22)$$

In particular, every noncyclic irreducible septic over a field of characteristic 29 has separating triple sums.

Proof. The primitive seventh root $\zeta = 23$ lies in the prime field, hence in K . Choose a Galois seven-cycle σ and number the roots by it. Suppose a collision gives $q(\sigma)x_0 = 0$, with $q = q_{A,B}$ as in (2). The nonzero-character part of q must have a zero: otherwise the Bézout argument in Theorem 3.3 would rule out the collision.

Table 1 shows that a pair with a resultant divisible by 29 is affine-equivalent to $A = 012$, $B = 346$. For the representative

$$q_*(T) = 1 + T + T^2 - T^3 - T^4 - T^6,$$

its values at the six nontrivial seventh roots are

$$(q_*(\zeta), q_*(\zeta^2), \dots, q_*(\zeta^6)) = (0, 14, 18, 25, 17, 20) \quad \text{in } \mathbb{F}_{29}.$$

There is exactly one zero. An affine change of the subset labels multiplies values by a nonzero root of unity and permutes the nontrivial characters, so every potentially colliding pair has exactly one such zero. Call its index j .

Project the relation $q(\sigma)x_0 = 0$ to the r th character by applying $\sum_{i=0}^6 \zeta^{-ir} \sigma^i$. Since σ fixes ζ , the result is

$$q(\zeta^r)u_r = 0.$$

Thus every nonconstant Fourier mode except possibly u_j vanishes. At least one is nonzero, so $u_j \neq 0$. Fourier inversion gives

$$x_i = \eta + \beta\zeta^{ij}, \quad \eta = \frac{u_0}{7}, \quad \beta = \frac{u_j}{7}.$$

The trace shows $\eta \in K$. The full root product is $(X - \eta)^7 - \beta^7$, and its coefficients show that $b = \beta^7 \in K$. This proves (i) $\Rightarrow$ (ii).

Conversely, all roots of a polynomial in (ii) are $\eta + \beta\zeta^i$. The characteristic-29 equality (5) persists after adding η to each root, since both sides contain three roots. Hence there is a triple-sum collision. The original polynomial is irreducible, and K contains μ_7 , so $K(\beta)$ is its cyclic degree-seven splitting field. Finally, translating each root adds 3η to each triple sum. The five-orbit calculation in Theorem 5.1, with t replaced by b , gives (22). $\square$

The conclusion is stronger than adding 29 to a blanket list of excluded characteristics. An exact solver in that characteristic can first test whether its input becomes a pure binomial after removing the trace. If it does, the equation has an immediate Kummer solution and the repeated-factor resolvent is understood completely. If it does not, the ordinary triple-sum resolvent separates. Cyclicity alone is not sufficient for a collision; the specific primitive element must give the translated binomial form.

11.2 Characteristic two: affine Fano configurations

A polynomial $L(Z) = \sum_i a_i Z^{2^i}$ in characteristic two is *additive*: $L(v + w) = L(v) + L(w)$. Its roots, when distinct, form a vector space over $\mathbb{F}_2$. This standard structure is exactly the one forced by a septic subset-sum collision.

Theorem 11.2 (Complete characteristic-two collision classification). *Let K have characteristic two, and let $f \in K[X]$ be monic, separable, and irreducible of degree seven. The following are equivalent:*

- (i) *Two different pairs of roots have equal sums.*
- (ii) *Two different triples of roots have equal sums.*
- (iii) *For some $\eta, a, b, d \in K$, with $d \neq 0$,*

$$f(X) = F(X - \eta), \quad F(Z) = Z^7 + aZ^3 + bZ + d. \quad (23)$$

In this case the roots are $\eta + (W \setminus \{0\})$, where W is a three-dimensional $\mathbb{F}_2$ -subspace of the splitting field. The center η is the trace of a root, and

$$R_{7,2,f}(Y) = F(Y)^3, \quad R_{7,3,f}(Y) = (Y - \eta)^7 f(Y)^4. \quad (24)$$

Its Galois group embeds in $\text{GL}_3(\mathbb{F}_2)$.

Proof. Adjoin a primitive seventh root ζ . This extends the base field by degree one or three, relatively prime to seven, so f remains irreducible: for a root field M/K of degree seven, the degree of $MK(\zeta)/K$ is divisible by both 7 and $[K(\zeta) : K]$ and is at most their product, forcing equality. Its splitting-field group over $K(\zeta)$ consequently contains a seven-cycle σ fixing ζ .

Assume (i) or (ii). The collision yields a nonzero polynomial $q \in \mathbb{F}_2[T]$ with $\deg q \leq 6$, $q(1) = 0$, and $q(\sigma)x_0 = 0$. In characteristic two,

$$\Phi_7(T) = (T^3 + T + 1)(T^3 + T^2 + 1),$$

with both cubic factors irreducible. The gcd of q and Φ_7 cannot be one, by the safe-direction proof of Theorem 3.3. It cannot be all of Φ_7 either: degree at most six would give $q = \Phi_7$, contradicting $q(1) = 0$ and $\Phi_7(1) = 1$. Therefore it is exactly one of the two cubic factors.

The primitive roots in either factor have indices $\{1, 2, 4\}$ or $\{3, 5, 6\}$. Reversing the cyclic numbering interchanges these sets. Character projection as in Theorem 11.1 therefore shows that, after such a reversal if necessary, the roots have form

$$x_i = \eta + v_1\zeta^i + v_2\zeta^{2i} + v_4\zeta^{4i}, \quad \eta = \frac{u_0}{7} = u_0 \in K.$$

On the eight-element subfield generated by ζ , the map

$$\psi(z) = v_1z + v_2z^2 + v_4z^4$$

is $\mathbb{F}_2$ -linear. It is injective: a nonzero kernel element h would make $\psi(z) = \psi(z + h)$ for distinct nonzero $z, z + h$, contradicting distinctness of the seven roots. Hence $W = \psi(\mathbb{F}_8)$ has dimension three, and the roots are exactly $\eta + (W \setminus \{0\})$.

For a finite $\mathbb{F}_2$ -vector space W in a field, its monic root polynomial $L_W(Z) = \prod_{w \in W} (Z - w)$ is additive. One elementary proof is induction on the dimension. If $W = V \oplus \mathbb{F}_2h$, then

$$L_W(Z) = L_V(Z)L_V(Z - h) = L_V(Z)^2 - L_V(h)L_V(Z),$$

which is additive whenever L_V is. Since $|W| = 8$ here,

$$L_W(Z) = Z^8 + aZ^4 + bZ^2 + dZ.$$

But $L_W(Z) = Zf(Z + \eta)$ belongs to $K[Z]$, so its displayed coefficients lie in K . Its distinct roots force $d \neq 0$. This proves (iii).

Conversely, under (iii), $ZF(Z)$ is an additive polynomial of degree eight with derivative $d \neq 0$. Its eight roots form a three-dimensional $\mathbb{F}_2$ -space W , and the nonzero roots are exactly the roots of F . For each nonzero $w \in W$, exactly three unordered pairs of distinct nonzero elements sum to w . Exactly seven unordered triples sum to zero, and exactly four sum to each nonzero w . These counts follow respectively by choosing one summand outside $\{0, w\}$, and by the seven Fano lines and the remaining 28 triples. After translation by η , pair sums are unchanged and triple sums acquire η . This proves both collisions and the identities (24).

Finally, every Galois automorphism fixes η and permutes W while preserving addition. Its action on W is faithful because the roots generate the splitting field. Thus the Galois group embeds in $\text{GL}_3(\mathbb{F}_2)$. $\square$

Corollary 11.3 (A nonsolvable colliding configuration). *There are separable irreducible septics in characteristic two with colliding pair and triple sums and with Galois group of order 168, isomorphic to $\text{GL}_3(\mathbb{F}_2)$.*

Proof. Let z_1, z_2, z_3 be algebraically independent over $\mathbb{F}_2$, put $L = \mathbb{F}_2(z_1, z_2, z_3)$, and let $G = \text{GL}_3(\mathbb{F}_2)$ act by invertible linear substitutions. Set $K = L^G$. The fixed-field theorem for a finite group of field automorphisms gives $\text{Gal}(L/K) = G$. The seven nonzero linear forms $v_1z_1 + v_2z_2 + v_3z_3$ with $v \in \mathbb{F}_2^3 \setminus \{0\}$ are distinct, form one G -orbit, and generate L . Their monic root polynomial is therefore separable irreducible of degree seven over K , with splitting field L , and its nonzero root set is an additive three-space minus zero. Theorem 11.2 applies.

The group has order $(8-1)(8-2)(8-4) = 168$. It is nonsolvable: it is generated by elementary transvections, and for distinct i, j, k each transvection $I + E_{ij}$ is the commutator of $I + E_{ik}$ and $I + E_{kj}$. Thus the nontrivial group is perfect and cannot be solvable. $\square$

The two exceptional characteristics therefore have markedly different geometry. In characteristic 29, a collision forces a single nonconstant Fourier character and a cyclic binomial equation. In characteristic two, a three-character Frobenius orbit survives and produces an affine Fano configuration, which can support a nonsolvable Galois group. The exceptional characteristic seven remains a separate unipotent problem, since its cyclic action cannot be diagonalized by seventh roots of unity.

12 A global obstruction: the universal cyclic cover is not one Kummer equation

The local pivot formulas work, but could a more ingenious single regular radical coordinate work on the entire distinct-root locus? The answer is negative in a precise algebraic sense. We prove it, rather than infer it from the zeros of a few familiar formulas.

12.1 The universal ordered configuration

Let k be algebraically closed of characteristic zero and contain a primitive p th root of unity. Put

$$A = k[x_0, \dots, x_{p-1}, \Delta^{-1}], \quad \Delta = \prod_{i < j} (x_i - x_j)^2. \quad (25)$$

Thus $\text{Spec } A$ parametrizes ordered tuples of pairwise distinct roots. Let $C_p = \langle \sigma \rangle$ act by cyclically permuting the variables, and write $B = A^{C_p}$.

No nonidentity cyclic shift fixes a geometric point: since p is prime, such a fixed point would have all coordinates equal. The quotient

$$\text{Spec } A \longrightarrow \text{Spec } B$$

is therefore a finite étale C_p -torsor. The term *torsor* means that a geometric fiber has a free transitive C_p -action; it is the algebraic version of a covering with that deck group.

A global regular Kummer presentation would be a C_p -equivariant isomorphism

$$A \cong B[T]/(T^p - a), \quad a \in B^\times, \quad (26)$$

where a generator acts by $T \mapsto \zeta^j T$ for some $j \neq 0$. The unit condition expresses the absence of ramification on the already separable locus. It is not an arbitrary strengthening: a Kummer equation with a zero radicand would fail to be étale at that zero.

The general Kummer framework distinguishes torsors given by a unit from those encoded by a line bundle whose p th tensor power is trivial [3]. We now compute the obstruction in this particular configuration explicitly.

12.2 The complete unit group

For $1 \leq a \leq (p-1)/2$ and $j \in \mathbb{F}_p$, define the oriented difference

$$\delta_{a,j} = x_j - x_{j+a}.$$

Every unordered pair occurs exactly once among these factors, up to a constant sign. Since the polynomial ring is a unique factorization domain, localizing at Δ gives

$$A^\times = \left\{ c \prod_{a=1}^{(p-1)/2} \prod_{j \in \mathbb{F}_p} \delta_{a,j}^{n_{a,j}} : c \in k^\times, n_{a,j} \in \mathbb{Z} \right\}. \quad (27)$$

The representation is unique once the chosen orientations are fixed. The cyclic action simply sends $\delta_{a,j}$ to $\delta_{a,j+1}$, without an additional sign. In multiplicative-module notation,

$$A^\times \cong k^\times \times \mathbb{Z}[C_p]^{(p-1)/2}. \quad (28)$$

Theorem 12.1 (No nontrivial eigenunit). *If $u \in A^\times$ and $\sigma(u) = \zeta^r u$, then $r = 0$ in $\mathbb{F}_p$. Consequently the universal cyclic cover has no presentation (26).*

Proof. Write u as in (27). Equality up to a nonzero constant does not change its exponents along the irreducible difference factors. Therefore $n_{a,j} = n_{a,j-1}$ for all a, j . For each a , the exponent is constant around its full cyclic orbit. The product $\prod_j \delta_{a,j}$ is itself invariant, and the constant c is fixed. Thus $\sigma(u) = u$. Since $u \neq 0$, this forces $\zeta^r = 1$.

In (26), the image of T would be a unit, because $T^p = a \in B^\times$, and would have a nontrivial eigencharacter. This is impossible by the first assertion. Even an initially unequivariant B -algebra presentation of this form would yield such an eigenunit: $\sigma(T)^p = T^p$ forces $\sigma(T)/T \in \mu_p$, and this character must be nontrivial because T generates the extension. $\square$

This rules out a single everywhere-regular Kummer coordinate on the cyclic quotient, even though rational Kummer coordinates exist over its function field. It does not rule out piecewise formulas, formulas on a dense open subset, or a nontrivial line-bundle formulation. Nor is it a claim about an unrestricted language of nested algebraic expressions with arbitrary case distinctions.

12.3 The Picard group of the cyclic quotient

Recall that $\text{Pic } B$ is the abelian group of isomorphism classes of invertible B -modules, with tensor product as its operation. Such modules are the algebraic counterparts of line bundles. Since A is a localization of a polynomial unique factorization domain, every invertible A -module is free of rank one; in particular, $\text{Pic } A = 0$.

Lemma 12.2 (Descent calculation). *For the above torsor,*

$$\text{Pic } B \cong H^1(C_p, A^\times).$$

Proof. An invertible B -module becomes free of rank one over A . Choose a generator after this base change. Transport of the generator under $g \in C_p$ is multiplication by a unit $a_g \in A^\times$. Compatibility of the action is exactly the cocycle equation $a_{gh} = a_g g(a_h)$. Changing the chosen generator changes the cocycle by a coboundary.

Conversely, a unit-valued cocycle gives a semilinear action on the free rank-one A -module. Faithfully flat descent along the finite étale torsor recovers an invertible B -module. These constructions are inverse and respect tensor products. The use of $\text{Pic } A = 0$ is precisely what permits one generator after base change for every class. $\square$

Theorem 12.3 (Picard group and the intrinsic chart obstruction). *For odd prime p ,*

$$\text{Pic}(k[x_0, \dots, x_{p-1}, \Delta^{-1}]^{C_p}) \cong \mathbb{Z}/p\mathbb{Z}. \quad (29)$$

The nonzero Fourier-character eigenspaces of A are invertible B -modules representing the nonzero powers of a generator of this group.

Proof. By (28) and Lemma 12.2, it suffices to compute first cohomology of the factors. The action on $k^\times$ is trivial, so a cocycle is just a character. Thus

$$H^1(C_p, k^\times) = \text{Hom}(C_p, k^\times) = \mu_p \cong \mathbb{Z}/p\mathbb{Z}.$$

For the additive regular module $M = \mathbb{Z}[C_p]$, a cocycle is determined by $v \in M$ satisfying

$$(1 + \sigma + \dots + \sigma^{p-1})v = 0.$$

In coordinates this says that the sum of the p coefficients of v is zero. Every such vector is a cyclic difference $(\sigma - 1)w$: solve for the coefficients of w by successive partial sums, whose consistency

around the cycle is exactly the zero-sum condition. Hence $H^1(C_p, \mathbb{Z}[C_p]) = 0$. Equation (29) follows.

For each $r \in \mathbb{F}_p$, let

$$L_r = \{a \in A : \sigma(a) = \zeta^r a\}.$$

A C_p -torsor becomes the regular permutation representation after a faithfully flat base change. Its character pieces therefore have rank one, and multiplication gives isomorphisms $L_r \otimes_B L_s \cong L_{r+s}$. Descent identifies these modules with the corresponding character cocycles, up to the harmless choice of which character is called a generator. Thus L_r has class r times that generator. For $r \neq 0$, the class is nonzero and has order p . $\square$

This also explains why p th powers can be invariant without their chosen p th roots fitting into one global regular coordinate. The obstruction is not a failure of Kummer theory; it is exactly the line-bundle alternative allowed by Kummer theory.

12.4 The Fourier charts trivialize the obstruction locally

Define universal modes u_j by (14) in A , and let $a_j = u_j^p \in B$. The basic opens

$$D(a_j) \subseteq \operatorname{Spec} B, \quad j = 1, \dots, p-1,$$

cover $\operatorname{Spec} B$. Indeed, a point missing all of them lifts to a geometric point of $\operatorname{Spec} A$ with every nonconstant Fourier mode zero; inversion would make all its coordinates equal, contradicting $\Delta \neq 0$.

On $D(a_j)$, u_j is a unit and is a generator of L_j . Every other character component can be divided by the corresponding power of u_j to become invariant. Consequently

$$A[a_j^{-1}] \cong B[a_j^{-1}][T]/(T^p - a_j), \quad T \mapsto u_j. \quad (30)$$

To see that there are no missing elements, decompose an element into its p character components by averaging with powers of ζ ; each component is an invariant multiple of one of $1, u_j, \dots, u_j^{p-1}$. These powers are independent by their distinct characters, and the monic relation is exactly $u_j^p - a_j$.

The overlap formulas (18)–(19) are the concrete local gluing data for this nontrivial torsor. Theorem 12.1 says that no single chart of the same global regular kind can replace the cover. Corollary 10.2 makes the narrower coordinate-cover bound sharp even on rational examples.

13 A practical proof architecture for ProveIt

The article suggests a modular extension rather than a monolithic new formula. Existing Fourier inversion can remain unchanged. Each added module should expose the assumptions needed by its consumers and avoid a statement whose hypotheses already contain the desired factorization.

[write] Status of this section. This section is a plan. None of it is formalized, and no Lean or Rocq code was delivered with the manuscript or is shipped with this report. Two facts about the tree should be weighed before acting on it. First, descriptor separation already exists for sextics in `Algebra/PolynomialFormulas/Lean/PolynomialFormulas/SexticSeparatingInvariants.lean` (note after Theorem 7.2); the proposed `elementarySubsetDescriptor_injective` and `separatingParameter_exists` would generalize it rather than start from nothing. Second, the arbitrary-degree Fourier core has a Rocq twin, `Algebra/PolynomialFormulas/Coq/LazardGeneralFourier.v` (note in Section 1), so the atlas cluster could be built on either prover. The plan proposes Lean only.

13.1 Separating the four certificates

A coefficient-level solver should keep four logically different objects:

- (1) **Root and Galois data:** a separable irreducible polynomial, its splitting field, and a justified permutation action on its roots.
- (2) **Separation data:** an injective equivariant invariant on the relevant finite set, or a proved squarefreeness test for its orbit product.
- (3) **Invariant recovery data:** exact equalities connecting computed coefficients to $u_0, a_j, c_{j,k}$ in a specified intermediate field.
- (4) **Branch data:** one nonzero pivot and one equation $\rho^p = a_j$; coherence of all reconstructed modes then follows from Theorem 8.1.

The second item is not implied by the first. The fourth need not contain $p - 1$ independent radical choices. These distinctions address two different ways an otherwise plausible solver can be unsound.

13.2 Suggested declaration-level milestones

The following are proposed names and statement boundaries, not purported existing Lean identifiers.

Proposed declaration	Essential conclusion
<code>subsetSum_injective_of_resultant_n e_zero</code>	A root p -cycle and nonzero cyclotomic resultants imply injectivity of k -subset sums.
<code>badCharacteristic_binomial_witness</code>	A common primitive cyclotomic root gives an Eisenstein binomial counterexample over a rational function field.
<code>septicTriple_badPrimes</code>	Finite determinant/orbit certificate yields exactly $\{2, 7, 29\}$.
<code>elementarySubsetDescriptor_injecti ve</code>	Subset elementary symmetric vectors distinguish subsets in every characteristic.
<code>separatingParameter_exists</code>	A union-of-root-sets argument proves the finite hitting bound.
<code>fourierPivot_ratios_fixed</code>	The pivot ratios and pivot power lie in the fixed field.
<code>fourierPivot_allBranches_shift</code>	Every branch of the single p th radical gives a cyclic shift of the original root tuple.
<code>fourierSupport_pureExtension_reali zation</code>	The rational family realizes each prescribed nonempty support.
<code>cyclicConfiguration_noEigenunit</code>	The explicit unit group has no nontrivial character unit.

The first cluster needs finite group actions, polynomial resultants, splitting fields, and finite sets. The atlas cluster needs only field algebra, finite sums, a primitive character, and a nonzero pivot. It is therefore a natural small next target, close to the inspected repository module. The global Picard calculation is conceptually stronger but needs substantially more algebraic geometry and descent infrastructure; it should not block the elementary solver improvements.

13.3 Eliminating hidden circularity

A theorem saying that a returned vector is correct *provided* it satisfies the full root factorization is not an invariant-recovery theorem. Likewise, testing numerical roots against a target polynomial

is not an exact construction of the intermediate-field coefficients. A useful audit should trace every exported root theorem backward to coefficient data, nonzero tests, and explicit field operations.

For the finite classification, a checker can accept integer matrices, verify their determinants by a fraction-free algorithm, verify affine orbit coverage, and infer the prime set. It should not trust a table solely because a computer algebra system printed it. The bundled auditor is an ordinary executable model of that boundary; a verified Lean or Rocq checker would be a separate deliverable.

For the branch theorem, no polynomial coefficient expansion is necessary. The short proof works for every odd prime at once. Keeping this theorem generic avoids large quintic- or septic-specific ring-normalization goals and makes exceptional support cases explicit rather than leaving them to a generic nonvanishing assumption.

14 Further research questions and topics

The questions below are proposed follow-on research problems. Their status as previously published open problems has not been asserted. Each has a specific target and a natural first test.

Research question 14.1 (Bad characteristics in higher prime degrees). Determine $\mathcal{B}_{p,k}$ for $p = 11, 13$ and the middle subset sizes. Can the affine-orbit computation be replaced by a structural description of the prime divisors of cyclotomic norms of balanced $\{0, \pm 1\}$ vectors?

Theorem 3.3 makes this a finite exact problem with guaranteed counterexamples for every reported bad prime. A useful first deliverable is an independently checked table, followed by explanations of unusually large prime divisors. Complementation, cancellation of common subset elements, and cyclotomic automorphisms should reduce the search.

Research question 14.2 (One bad prime, different Galois groups). For higher p and $\ell \in \mathcal{B}_{p,k}$, which transitive Galois groups can occur for colliding examples? Can the characteristic-2 and characteristic-29 rigidity classifications for septics be extended to a complete classification in characteristic seven?

Theorems 11.1 and 11.2 already answer two of the septic cases: characteristic 29 forces a cyclic binomial configuration, whereas characteristic two permits a nonsolvable group of order 168. For higher primes, the number and arrangement of common cyclotomic roots should impose analogous Fourier-support constraints. Characteristic p instead requires a unipotent cyclic-operator analysis.

Research question 14.3 (Smaller deterministic separating families). For $(n, k) = (7, 3)$, replace the bound 1191 by a substantially smaller uniform number for a specified family of separating invariants. Can one exploit the Galois action to count entire collision orbits at once?

The present bound simply adds the degrees of all pair differences, often counting the same exceptional parameter many times. It is not claimed to be sharp. A good improvement must still handle finite characteristic and must distinguish the existence of one good parameter from a fixed short hitting set over the original field.

Research question 14.4 (An exact characteristic audit for matching predicates). For the additional root relations used to match invariant factors in septic formulas, determine the corresponding balanced or unbalanced coefficient vectors and their cyclotomic obstructions. Is there one finite set of excluded characteristics that certifies every matching step?

This is more precise than assuming that the prime divisors of $p(p-1)$ are the only possible problems. The characteristic-29 calculation shows that auxiliary invariants can introduce other primes. Unbalanced relations require separately treating their constant-character component.

Research question 14.5 (Minimal nonlinear covers of the cyclic configuration). What is the smallest number of regular Kummer charts needed to cover $\text{Spec } B$ when arbitrary nontrivial character sections are allowed?

We prove that one global chart is impossible and that the standard Fourier coordinate cover needs all $p-1$ of its members. These facts do not settle the optimum among nonlinear choices. This question links radical formulas to the number of generators of the nontrivial invertible modules L_j .

[write] **Answered in Part II, 29 September 2026 (batch 52)**. The minimum is $d(p-1)$ charts for $\text{Conf}_p(\mathbb{A}^d)/C_p$, hence $p-1$ for the question as posed ($d=1$): four for quintics and six for septics (Corollary 17.2). It comes from the integral Chow ring

$$\text{CH}^*(\text{Conf}_p(\mathbb{A}^d)/C_p) = \mathbb{Z}[t]/(pt, t^{d(p-1)}), \quad t = c_1(\mathcal{L})$$

(Theorem 17.1). A cover by m regular Kummer charts trivializes the character line on every chart and so forces $t^m = 0$ (Lemma 21.1), while $t^{d(p-1)-1} \neq 0$; the $d(p-1)$ Fourier charts attain the bound (Proposition 21.3). The charts may be arbitrary Zariski opens with different nontrivial characters. Each nontrivial eigenmodule L_j needs exactly $p-1$ generators as an invertible B -module, which settles the link named in the last sentence above. Part II works over any field of characteristic different from p containing ζ , so it covers this section's algebraically closed field of characteristic zero. Part II is not formalized.

Research question 14.6 (Other permutation subgroups and their unit characters). Compute $\text{Pic}(A^H)$ and its radical-coordinate obstructions for other subgroups $H \leq S_n$ acting freely on ordered distinct-root configurations, beginning with the affine prime-degree groups.

The descent method still reduces the problem to $H^1(H, A^\times)$. However, permutations can reverse orientations of difference factors, so the clean cyclic splitting (28) can acquire character corrections. Those signs must be derived rather than discarded.

Research question 14.7 (Coefficient-efficient invariant ratios). For quintics and septics, construct small exact expressions for the chart data $a_j, c_{j,k}$ in a computationally convenient presentation of the abelian intermediate field, and compare them with existing large invariant formulas.

The atlas theorem guarantees that only one p th radical is necessary once these invariants are known. The unresolved algorithmic issue is computing them without already knowing an ordered root tuple. Modular reconstruction, trace pairings, and carefully chosen resolvents are plausible approaches.

Research question 14.8 (Verified finite certificates and branch reconstruction). Can the 29 norm determinants, the affine orbit partitions, and the generic one-radical chart theorem be formally verified with a small trusted kernel boundary and without native computation assumptions?

The provided certificates are deliberately small. A realistic first milestone is the septic triple classification and the one-radical theorem, followed by a coefficient-level example and a full audit of its dependency graph. This is a formalization target, not a claim that such a proof is already included.

Research question 14.9 (Numerically stable exact chart selection). Among nonzero Fourier charts, which selection criterion minimizes intermediate coefficient growth and numerical conditioning while remaining compatible with exact certification?

The algebraic theorem only distinguishes zero from nonzero. A practical hybrid solver could use approximate information to propose a pivot and exact arithmetic to certify it. The mathematical output must remain independent of the heuristic and preserve the transition identities.

15 Conclusion

Two kinds of exceptional behavior have different causes. Subset-sum collisions are arithmetic: they are controlled exactly, away from the prime degree itself, by cyclotomic norm divisibility. The exceptional characteristic 29 for septic triple sums is a concrete manifestation, with a separable irreducible cyclic witness and a complete repeated-factor resolvent. The collision locus is even rigid: characteristic 29 forces a translated pure binomial, while characteristic two forces an affine Fano configuration and an additive-polynomial quotient. A separating vector compressed to a scalar supplies a uniform repair with an explicit finite search bound.

Exceptional radical coordinates are geometric as well as arithmetic. Every nonempty Fourier support occurs even for rational examples with full affine Galois group. A nonzero pivot gives a coherent one-radical local formula, but the universal cyclic cover carries a nontrivial order- p line-bundle class and admits no single global regular Kummer generator. The right object is therefore an atlas with certified transitions, not a formula whose denominators are assumed never to vanish.

For ProveIt, the immediate gain is a sharper proof architecture: separate root separability from resolvent separation, separate invariant recovery from Fourier inversion, and derive branch coherence from one shared radical. The results here supply exact mathematical statements and small certificates for those interfaces, while leaving coefficient optimization and kernel formalization as explicit next tasks.

A Reproducibility and the finite proof boundary

A.1 Files in the archive

The archive contains this source and compiled PDF, a README, two checking programs, and generated certificate files. The principal program is `verify_results.py`; it uses exact SymPy integer and rational arithmetic, not floating-point roots. The independent program `audit_certificates.py` uses only the Python standard library to check the orbit and determinant certificates.

Typical commands are

```
python -m pip install sympy
python verify_results.py --out certificates
python audit_certificates.py --cert certificates/cyclotomic_norm_certificates.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The source uses common LaTeX packages and has no external figure or bibliography-file dependency.

[write] Shipped names. The commands above use the delivery layout. In this report the scripts are `code/verify_results.py` and `code/audit_certificates.py`, the certificate files are in `data/` (not `certificates/`), and the delivered PDF is not shipped. Run from the report directory, writing fresh outputs to a scratch directory so that the shipped records are never overwritten:

```
uv run --no-project --with sympy==1.14.0 python code/verify_results.py --out <tmpdir>
```

```
py code/audit_certificates.py --cert data/cyclotomic_norm_certificates.json
```

The auditor is read-only without `--out` and prints its report, which equals `data/independent_audit_report.json`. At the intake the fresh outputs of the verifier equalled the files in `data/` after removing carriage returns; the "python" field of `verification_report.json` records the interpreter and may differ.

A.2 Recorded checks

The executed full verification run used Python 3.13.5 and SymPy 1.14.0. The report `certificates/verification_report.json`² records a passing result for:

Check	Count
Exact pairwise cyclotomic resultants	850
Independent orbit-representative integer determinants	29
Finite-field counterexample families	4
Distinct repaired characteristic-29 coefficient vectors	35
Nonempty Fourier supports, degrees five and seven	78
Exact Fourier-mode identities	516
Nonzero pivot charts	224
Reconstructed root entries over all branches in those charts	10,208
Exact worked minimal polynomials	6
Solvable affine septic orbit patterns	4
Finite-field character-root patterns for septic rigidity	1,400

The Fourier test algebra is

$$\mathbb{Q}[\alpha, \zeta]/(\alpha^p - 2, \Phi_p(\zeta)).$$

Its elements are stored with exact rational coefficients in the basis $\alpha^a \zeta^b$, $0 \leq a < p$, $0 \leq b < p - 1$. The field degree calculation in Theorem 10.1 justifies this basis. The tests run through every nonempty support with coefficient one, every available pivot, all p choices of the radical branch, and all p reconstructed roots. General rational coefficients are covered by the proof, not exhaustively by this finite test.

The checks do not prove that the cited repository modules compile in a fresh environment, and no Lean or Rocq process was run for this article. Nor do these checks verify the Picard calculation: that part is the written descent proof in Section 12. Keeping these boundaries explicit prevents an exact computation from being mistaken for an unrelated formal verification claim.

B A compact dependency map

The central results have the following logical dependencies:

Cauchy theorem + cyclic operator Bézout identity

$\implies$ safe resultant criterion,

cyclotomic common root + Eisenstein binomial

$\implies$ necessity at every non- p bad prime,

finite norm tables + exact criterion + Artin–Schreier witness

²[write] Shipped in this report as `data/verification_report.json`.

$\implies$ complete small-degree classification,
 septic norm table + characteristic-29 character projection
 $\implies$ translated-binomial rigidity,
 two cubic cyclotomic factors in characteristic two
 $\implies$ affine Fano and additive-quotient rigidity,
 subset polynomial identity + finite root bound
 $\implies$ uniform separating repair,
 Fourier covariance + one nonzero mode + inversion
 $\implies$ all-branch coherent reconstruction,
 prime-degree pure extension + geometric sums
 $\implies$ all support patterns,
 factorial configuration ring + explicit units + descent
 $\implies \text{Pic } B \cong \mathbb{Z}/p\mathbb{Z}$ and no global eigenunit.

There is no dependence of the arithmetic classification on the geometric Picard theorem, and no dependence of the Fourier chart theorem on the finite norm table. The three strands can be independently reviewed and formalized.

Part II. Optimal Kummer atlases for prime cyclic configurations

Sections 1–15 and Appendices A–B form the original report (Part I), unchanged except for dated [write] notes pointing to this addition. Part II was added on 29 September 2026 in batch 52 of ProveIt’s incoming-report intake, from one manuscript written against Part I as printed above. It answers Part I’s Research question 14.5: covering the cyclic quotient $\text{Spec } B$ of the distinct-root configuration by regular Kummer charts, with arbitrary nonlinear character sections allowed, needs exactly $p - 1$ charts. More generally, $\text{Conf}_p(\mathbb{A}^d)/C_p$ has integral Chow ring $\mathbb{Z}[t]/(pt, t^{d(p-1)})$ and needs exactly $d(p - 1)$ charts. Section 16 records the provenance, fixes the notation and states the formal status of the addition; Sections 17–29 and Appendix C are the manuscript.

PROVEIT RESEARCH COMPANION

Optimal Kummer Atlases for Prime Cyclic Configurations

Exact nonlinear chart complexity, torsion Chow classes,
and prime-degree radical reconstruction

Research prepared for Vladimir Reshetnikov

29 September 2026

AI-assisted research manuscript; written proofs, not Lean/Rocq formalization.

Abstract (manuscript 09). The ProveIt report *Specialization-Safe Radical Solvers* asks whether arbitrary nonlinear character sections can cover the universal cyclic root configuration with fewer regular Kummer charts than the standard $p - 1$ Fourier charts. We answer this question: the exact minimum is $p - 1$, even when the covering opens are arbitrary and the nontrivial character may vary between charts.

More generally, let p be prime, let $d \geq 1$, and let k have characteristic different from p and contain a primitive p th root of unity. For the free cyclic quotient $X_{p,d} = \text{Conf}_p(\mathbb{A}_k^d)/C_p$, write $\mathcal{L}$ for a generating character line bundle and $N = d(p - 1)$. We prove

$$\text{CH}^*(X_{p,d}) = \mathbb{Z}[t]/(pt, t^N), \quad t = c_1(\mathcal{L}).$$

The exact regular Kummer-chart count and the exact number of global sections generating any nontrivial character line bundle are both N . The proof identifies the contribution of the full diagonal and proves that deleting all proper partial diagonals introduces no further Chow relations. It uses integral cycle classes; rational coefficients would erase the obstruction.

Further consequences include sharp noncompression for nonlinear character covariants, explicit nonzero torsion cycles represented by pure Fourier-mode loci, and invariance under allowing arbitrary partial collisions while excluding the full diagonal. In a different measure of complexity, the finite algebra of the cyclic cover needs exactly d global algebra generators. Thus in the original one-dimensional problem the cover is globally monogenic as an ordinary polynomial algebra, even though its regular Kummer atlas needs $p - 1$ charts. Exact finite checks, a proof-dependency audit, and concrete further research problems accompany the proofs. The resolution of the named repository question is explicit; priority over the entire literature is not asserted.

Main answer in the original setting. For odd prime degree p , nonlinear regular Kummer charts cannot improve on the Fourier cover. The optimum is $p - 1$: four charts for quintics and

six for septs. This is a statement about a universal cyclic quotient, not a count of radicals needed for one polynomial.

16 [write] Provenance, conventions and status of Part II

16.1 One manuscript, one addition

Part II is batch-52 manuscript 09, *Optimal Kummer Atlases for Prime Cyclic Configurations: Exact nonlinear chart complexity, torsion Chow classes, and prime-degree radical reconstruction*, an AI-assisted research manuscript prepared for Vladimir Reshetnikov and dated 29 September 2026 (22 pages, A4, as delivered). It arrived as the archive `ProveIt_Optimal_Kummer_Atlases.zip` (inner directory `Optimal_Kummer_Atlases/`, main file `optimal_kummer_atlases.tex`) in ProveIt commit 458ccfb80 and was placed in a701d9098. It was written against ProveIt commit 1ee53d57de253d16cdaad79d1f54bbd95d76d682, at which Part I’s `article.tex` had the Git blob f8303298d0. That blob was unchanged at the placement commit, so every statement the manuscript makes about “the report” refers to Part I as printed above, before the dated notes of this write. Its secondary repository source, `Algebra/PolynomialFormulas/LazardQuinticFormalization.tex` (blob 5281cbf260), is also unchanged. The manuscript’s checker, build script and three JSON records are shipped beside Part I’s files with the prefix `02-kummer-atlases-` (Part I’s unprefixed files count as 01); its source, README and PDF are not shipped and survive in the arrival commit. The manuscript did not see any other batch-52 delivery.

The manuscript contributes, in the order printed here: the question, the main theorem and its scope (Section 17); character lines and the equivalence between Kummer charts and trivializations (Section 18); the Chow ring of the quotient with only the full diagonal removed (Section 19); invariance under deleting the partial diagonals (Section 20); the exact chart count (Section 21); eigenline generators and nonlinear covariant noncompression (Section 22); local reconstruction with one shared radical in every dimension d (Section 23); the exact ordinary algebra-generator number d and configuration noncompression (Section 24); invariance under allowed partial collisions (Section 25); explicit torsion cycles and small cases (Section 26); a proof audit, its finite checks and a formalization decomposition (Section 27); eight research questions (Section 28); its conclusion (Section 29); and its provenance and literature comparison (Appendix C). Every result, proof, example, remark, table, limitation and question of the manuscript is printed.

Where the merge had to choose:

- **Numbering.** The manuscript’s Section n is Section $n + 16$ here and its Appendix A is Appendix C; theorem-like numbers follow the sections (the manuscript’s Theorem 1.1 is Theorem 17.1). Equation numbers continue Part I’s: the manuscript’s equation (m) is printed as equation $(m + 30)$.
- **Front matter.** The manuscript’s title block, abstract and “Main answer” paragraph open Part II; its table of contents is replaced by the report’s.
- **Overlap with Part I.** Three results of Part II restate or reprove Part I in dimension one, and all are kept. The local reconstruction (Section 23) is Part I’s Theorem 8.1 and branch transition for every d . The group $\mathrm{CH}^1 = \mathrm{Pic} \simeq \mathbb{Z}/p\mathbb{Z}$ for $d = 1$ is Part I’s Theorem 12.3, proved here by a different route. The Fourier charts and their covering property (Proposition 21.3) extend Part I’s (30) to every d . Dated notes at those places say so.
- **Bibliography.** Both sources cite the same Stacks Project section on Kummer theory; it is printed once, with both annotations. The manuscript’s other five entries keep their text under keys prefixed `ka:`. Its entry for “the report” is Part I at the pin.

- **Part I** is unchanged except for the title-page provenance line, a sentence in Section 1.5, an item in Section 1.6, a label on Research question 14.5, dated notes after Corollary 10.2 and after that question, a table-of-contents line for Part I, and the merged Stacks Project entry.

16.2 Conventions and notation

No symbol of the manuscript was renamed and no normalization was changed. Part II uses Part I's Fourier convention: $\sigma(x_{a,i}) = x_{a,i+1}$ and $u_{a,j} = \sum_i \zeta^{-ij} x_{a,i}$ (54), so for $d = 1$ the mode $u_{1,j}$ is Part I's u_j (14), with $\sigma(u_j) = \zeta^j u_j$ in both. The character line $\mathcal{L}_j$ is the sheaf of ζ^j -eigenfunctions, so for $d = 1$ its module of global sections is Part I's L_j (Theorem 12.3). A *regular Kummer chart* (Section 17) has a unit radicand and a nontrivial character, as in Part I's (26) and (30), but its open set is arbitrary, not necessarily principal or affine. Several letters carry more than one meaning inside Part II or between the two parts:

Symbol	Meaning in Part II	Not to be confused with
A, B, k	For $d = 1$: $A = k[x_0, \dots, x_{p-1}, \Delta^{-1}]$, $B = A^{C_p}$ (31); k any field with $\text{char } k \neq p$ containing ζ (32)	The same A, B in Section 12, where k is algebraically closed of characteristic zero; A, B as index subsets in Sections 2–7.
$X, X_{p,d}; Y$	The quotient $U_{p,d}/C_p$ (for $d = 1$, $X = \text{Spec } B$); $Y = U_0/C_p$, only the full diagonal removed	Part I's polynomial variable X and resolvent variable Y .
t	$c_1(\mathcal{L}) \in \text{CH}^1(X)$	The transcendental t of $X^7 - t$ over $\mathbb{F}_{29}(t)$ (Section 5).
N	$d(p - 1)$	Part I's norm $N_{A,B}$, $N = \binom{n}{k}$ and a minimal normal subgroup.
F	The full diagonal $\{z_0 = \dots = z_{p-1}\}$	The finite fields $\mathbb{F}_q$.
$a; a_{a,j}$	A coordinate index $1 \leq a \leq d$; the radicand $a_{a,j} = u_{a,j}^p$ of the chart $D_{a,j}$; a unit radicand a in $T^p - a$; a Bézout integer	Part I's $a_j = u_j^p$ (for $d = 1$, $a_{1,j} = a_j$).
$c_{a',k}; c$	$u_{a',k}/u^{e_j(k)}$ for a fixed pivot $u = u_{b,j}$, first index a coordinate (59); c is also a coordinate index and a ratio in the pivot change, and a Wilson coefficient in Section 19	Part I's $c_{j,k}$, first index the pivot. True: for $d = 1$ Part II's $c_{1,k}$ is Part I's $c_{j,k}$. False: it is Part I's $c_{1,k}$.
κ, γ, α	Least number of regular Kummer charts; least number of generating global sections; least number of global algebra generators	Part I's $\alpha = 2^{1/p}$ (Section 10, Appendix A).
q	The quotient map $U \rightarrow X$	Part I's $q_{A,B}$ (2); field sizes q in both checkers.
$E, E_M, \mathcal{E}_W$	A vector bundle; $\mathbb{A}^M \setminus \{0\}$; the bundle of the representation W	Part I's field E and descriptor $E(A)$.
W, W_π	An open set; the representation (43); $V_\pi \setminus F$	Part I's L_W (Section 11).
Z	Z_M , the cycles Z_S , and W/C_p in Theorem 25.1	$\mathbb{Z}$; Part I's elimination variable Z .
C, C_b	The deleted locus $Y \setminus X$ and its filtration	The group C_p .
h	The hyperplane class of $\mathbb{P}^{M-1}$; a pivot index in (62)	Part I's h in (18), the same role.
S	A smooth variety; strata S_π ; an index set in Proposition 26.1	Part I's Fourier support S (Theorem 10.1); Lazard's S_k .
f, M	Covariants f_i , a morphism f , and $f(T) = \prod(T - x_i)$; a line bundle M and the dimension M of E_M	Part I's input polynomial f and matrices M_q .
CH	Integral Chow groups throughout	Rational Chow groups, which vanish in positive degree here (Section 20).

Two tempting false readings are ruled out in the text itself and are repeated here. First, a nonzero class in $\text{Pic } B$ alone gives a lower bound of two charts, not $p - 1$; the higher powers $t^2, \dots, t^{p-2}$ are needed. Second, a cover by $p - 1$ charts does *not* mean that $p - 1$ independent p th radicals are needed at one input: each chart uses one radical (Section 23).

16.3 Formal status and what is not claimed

- **Not formalized.** No statement of Part II is formalized in Lean or Rocq, and no `ka:` label has a Lean or Rocq counterpart. The repository has no Chow-group development. The nearest project declaration is the Rocq lemma `cyclic_kummer_generator` (`Algebra/PolynomialFormulas/Coq/LazardCyclicKummerGenerator.v`, l. 32): a cyclic Galois extension containing the needed roots of unity has one Kummer generator. It is a field-level existence statement; Part II does not rely on it, and it says nothing about regular charts on the configuration space. Section 27’s formalization decomposition is a plan of proposed milestones, not existing declarations, and no Lean or Rocq code was delivered or is shipped.
- **Evidence.** The geometric results rest on written proofs that use standard integral intersection theory [16, 14, 15]. The finite checker verifies partition counts, Euler coefficients and Fourier identities only; it does not verify Chow groups, localization, or the atlas lower bound. Its Fourier tests are exhaustive over $\mathbb{F}_7$ and $\mathbb{F}_{11}$ and seeded samples over $\mathbb{F}_{29}$, and its branch checks cover 128 inputs per field.
- **The source’s own non-claims** stand as printed: the Chow-theoretic machinery is established work, and Totaro’s treatment of cyclic products is close background; no exhaustive priority review was made. Part II does not solve the generic quintic over its symmetric coefficient field, does not compute coefficient-level invariants, does not minimize nested radicals for one input, and gives no lower bound for programs with case distinctions. It concerns regular Kummer presentations on Zariski opens of the specified universal cyclic quotient. Theorem 25.1 concerns intermediate opens containing all distinct configurations, not further deletions, and the algebra-generator count is not asserted for those larger opens. The algebraic chart count is not claimed to equal holomorphic or continuous counts. The research questions of Section 28 are not asserted to be known open problems.
- **Review.** AI-assisted and unrefereed. At the write the proofs were read; the checker was rerun on a copy (Appendix C); and a separate script recomputed the proper-partition orbit counts from Stirling numbers, the Wilson residues $((p - 1)!)^d \equiv (-1)^d$, the integral identity behind (67) for $p \leq 13$, the quintic coefficient $2 \cdot 3 \cdot 4 \equiv -1 \pmod{5}$, the tuple and branch counts of the tables in Section 27, and the pivot-change shift $reh^{-1} \equiv rj^{-1}$. This is not an independent proof review.

17 The question, the answer, and the scope

17.1 The repository problem

The starting point is Research Question 14.5, entitled *Minimal nonlinear covers of the cyclic configuration*, in the ProveIt research report *Specialization-Safe Radical Solvers* [12]. Its setting is

$$A = k[x_0, \dots, x_{p-1}, \Delta^{-1}], \quad \Delta = \prod_{i < j} (x_i - x_j)^2, \quad B = A^{C_p}, \quad (31)$$

where p is odd prime and a generator σ cyclically permutes the variables. The report works over an algebraically closed field of characteristic zero for this geometric question. It establishes a nontrivial character line bundle, rules out a single global regular Kummer coordinate, and shows

that none of the $p - 1$ standard Fourier coordinate charts can simply be omitted. It leaves open the optimum when nonlinear character sections are allowed.

The distinction is genuine. Showing that every member of one particular cover is necessary does not show that a different cover cannot be smaller. Nor does a nonzero class in $\text{Pic}(B)$ alone give a lower bound larger than two. We determine the higher self-intersections of that class and use them to rule out *all* smaller covers.

The report is associated with the repository's polynomial-formula project and its audit of solvable-quintic reconstruction [13]. Our argument does not assume the correctness of any unformalized claim there: the Fourier construction, the line-bundle interpretation, and the needed obstruction are proved below.

[write] **The report cited here is Part I.** The report [12] is Part I of this report, cited at the pinned blob, which is the text of Part I printed above apart from the dated [write] notes of batch 52. Its Research Question 14.5 is Research question 14.5; the Sections 8, 10 and 12 named in the bibliography entry are Sections 8, 10 and 12. The nontrivial character line bundle is Theorem 12.3, the impossibility of one global regular Kummer coordinate is Theorem 12.1, and the irredundancy of the $p - 1$ Fourier coordinate charts is Corollary 10.2.

17.2 Notation and the main theorem

Fix throughout a prime p , an integer $d \geq 1$, and a field k satisfying

$$\text{char}(k) \neq p, \quad \zeta \in k, \quad \text{ord}(\zeta) = p. \quad (32)$$

Put

$$\begin{aligned} V_{p,d} &= (\mathbb{A}_k^d)^p, \\ U_{p,d} &= \text{Conf}_p(\mathbb{A}_k^d) = \{(z_0, \dots, z_{p-1}) : z_i \neq z_j \text{ for } i \neq j\}, \\ X_{p,d} &= U_{p,d}/C_p, \quad N = d(p-1). \end{aligned} \quad (33)$$

All inequalities in the configuration definition are scheme-theoretic open conditions, not restrictions to k -rational points. The action is cyclic permutation of the labels. The quotient map is denoted q . Let $\mathcal{L}_j$ be its weight- j character line bundle, so $\mathcal{L}_j \otimes \mathcal{L}_h \simeq \mathcal{L}_{j+h}$ and $\mathcal{L}_0 = \mathcal{O}_X$. We write $\mathcal{L} = \mathcal{L}_1$ and $t = c_1(\mathcal{L})$.

A *regular Kummer chart* is an open subset $W \subset X$ on which $q_*\mathcal{O}_U$ has an equivariant presentation

$$(q_*\mathcal{O}_U)|_W \simeq \mathcal{O}_W[T]/(T^p - a), \quad a \in \Gamma(W, \mathcal{O}_W^\times), \quad \sigma(T) = \zeta^j T,$$

for some $j \in \{1, \dots, p-1\}$. Opens need not be principal or affine. The number of charts in a smallest such finite cover is $\kappa(X)$. For a globally generated line bundle M , let $\gamma(M)$ be the least number of global sections generating it at every point.

Theorem 17.1 (Exact ring and exact nonlinear chart complexity). *Under (32), there is an isomorphism of graded integral rings*

$$\mathrm{CH}^*(X_{p,d}) \simeq \mathbb{Z}[t]/(pt, t^{d(p-1)}), \quad \deg t = 1, \quad (34)$$

carrying t to $c_1(\mathcal{L})$. Moreover,

$$\kappa(X_{p,d}) = \gamma(\mathcal{L}_j) = d(p-1) \quad (1 \leq j \leq p-1). \quad (35)$$

The chart lower bound permits arbitrary nonlinear regular character sections, arbitrary covering opens, and different nontrivial characters on different opens.

The ring calculation is proved in Sections 19 and 20; the chart and generator assertions are proved in Sections 21 and 22. When $N > 1$, the class t^r has exact order p for every $1 \leq r < N$. When $N = 1$, necessarily $(p, d) = (2, 1)$, the Chow ring is $\mathbb{Z}$ and a single chart works. Including this boundary case prevents an incorrect blanket assertion that every character line bundle here is nontrivial.

Corollary 17.2 (Answer to the repository question). *In (31), the minimum number of regular Kummer charts covering $\mathrm{Spec} B$ is $p - 1$. Each nontrivial character eigenspace of A requires exactly $p - 1$ generators as an invertible B -module.*

Proof. For $d = 1$, the ordered configuration is affine with coordinate ring A , and the finite quotient is $\mathrm{Spec} B$. Apply Theorem 17.1 and the equivalence between vector bundles on an affine scheme and finite projective modules. $\square$

17.3 What is and is not claimed

The mathematical contribution is a complete written answer to the named repository question, together with the higher-dimensional ring, noncompression, deletion-invariance, and generator statements proved here. The intersection-theoretic machinery is established work of Fulton, Totaro, and Edidin–Graham; our use of it is not a new foundation for Chow theory [14, 15]. Totaro’s treatment of cyclic products is particularly close background. A literature search and comparison with those sources do not constitute an exhaustive priority review, and no claim of first discovery of every intermediate formula is made.

The article does not solve a generic quintic over its symmetric coefficient field. It does not compute coefficient-level invariants, minimize the number of nested radicals for an individual input, or give a lower bound for arbitrary programs with case distinctions. It treats *regular Kummer presentations on Zariski opens of a specified universal cyclic quotient*. All main assertions have written proofs; none is asserted to have been checked by Lean or Rocq. The accompanying Python checks test finite combinatorics and Fourier identities only.

18 Cyclic torsors, eigenlines, and regular Kummer charts

18.1 The free locus and its quotient

Let $F \subset V = V_{p,d}$ be the full diagonal:

$$F = \{z_0 = z_1 = \cdots = z_{p-1}\} \simeq \mathbb{A}_k^d.$$

If a nonidentity element of C_p fixes a geometric point of V , it generates the entire group, and all the p coordinates of that point are equal. Thus the action is free on

$$U_0 = V \setminus F,$$

and in particular on $U = U_{p,d}$. The quotients $Y = U_0/C_p$ and $X = U/C_p$ are smooth quasiprojective k -schemes. The quotient maps are finite étale C_p -torsors of degree p . These quotients can be constructed as opens of the finite affine quotient V/C_p ; freeness makes the restrictions torsors. Smoothness descends along the étale quotient maps.

Here a torsor means that every geometric fiber has a free transitive C_p -action. It does *not* mean that this finite cover is trivial in the Zariski topology.

18.2 The character algebra

Because p is invertible and k contains ζ , the projectors

$$e_j = \frac{1}{p} \sum_{r=0}^{p-1} \zeta^{-jr} \sigma^r \quad (j \in \mathbb{F}_p) \quad (36)$$

act on $q_*\mathcal{O}_U$ and yield

$$q_*\mathcal{O}_U = \bigoplus_{j \in \mathbb{F}_p} \mathcal{L}_j, \quad \mathcal{L}_j = \{f : \sigma(f) = \zeta^j f\}. \quad (37)$$

After a faithfully flat base change trivializing the torsor, this is the character decomposition of the algebra of functions on C_p . Each summand has rank one, and multiplication gives isomorphisms

$$\mathcal{L}_j \otimes \mathcal{L}_h \xrightarrow{\sim} \mathcal{L}_{j+h}. \quad (38)$$

These properties descend. In particular, $\mathcal{L}^{\otimes p} \simeq \mathcal{O}_X$, $\mathcal{L}_j \simeq \mathcal{L}^{\otimes j}$, and $pt = 0$. The same constructions apply on Y and restrict to those on X .

Lemma 18.1 (A Kummer chart is a trivialization of a character line). *For a nonempty open $W \subset X$, the following are equivalent:*

- (i) W is a regular Kummer chart.
- (ii) $\mathcal{L}_j|_W$ is trivial for some $j \neq 0$ in $\mathbb{F}_p$.
- (iii) $\mathcal{L}|_W$ is trivial.

If these conditions hold, every nontrivial character line is trivial on W .

Proof. In a Kummer presentation, T generates the relevant eigenline, proving (i) $\Rightarrow$ (ii). If j is nonzero modulo p , choose integers a, b with $aj + bp = 1$. Using tensor powers and duals,

$$\mathcal{L} \simeq \mathcal{L}_j^{\otimes a} \otimes (\mathcal{L}^{\otimes p})^{\otimes b}.$$

Thus (ii) implies (iii), which also trivializes every $\mathcal{L}_j$.

Conversely, suppose s is a generator of $\mathcal{L}_j|_W$, where $j \neq 0$. The multiplication isomorphisms show that $s^p = a$ is a unit of $\mathcal{O}_W$. The powers $1, s, \dots, s^{p-1}$ generate all p distinct character pieces. They therefore give the asserted isomorphism $\mathcal{O}_W[T]/(T^p - a) \simeq (q_*\mathcal{O}_U)|_W$. $\square$

This is the concrete form of the standard Kummer description by an invertible sheaf and a trivialization of its p th tensor power [3]. Notice that a trivial eigenline does not require a to have a p th root on W . The finite torsor can remain nontrivial on a perfectly valid Kummer chart.

Remark 18.2 (Why the radicand must be a unit). If a vanishes at a point, the fiber $T^p - a$ has a multiple root over an algebraic closure. It cannot present an étale cover there. Unit radicands are consequently forced by the intended geometry rather than added as an artificial restriction.

Remark 18.3 (Equivariance does not hide a simpler one-equation chart). All nonempty opens of X are integral. Suppose the same cover has a W -algebra presentation $\mathcal{O}_W[T]/(T^p - a)$ with a a unit, without a specified action on T . On the integral inverse image of W , $\sigma(T)/T$ is a p th root of unity, hence one of the constant ζ^j . It cannot be 1, since then T would be invariant and could not generate a degree- p algebra. Thus such a presentation automatically has a nontrivial character, and is included in our chart definition.

19 The punctured cyclic quotient and its Chow ring

19.1 The intersection-theoretic input

We use integral Chow groups: $\mathrm{CH}_i(S)$ consists of i -dimensional algebraic cycles modulo rational equivalence. For smooth pure-dimensional S , put $\mathrm{CH}^r(S) = \mathrm{CH}_{\dim S - r}(S)$, with its intersection product. The needed properties are the following standard ones [16, 14, 15]: proper pushforward; pullback between smooth varieties; restriction to opens; the localization sequence

$$\mathrm{CH}_i(Z) \longrightarrow \mathrm{CH}_i(S) \longrightarrow \mathrm{CH}_i(S \setminus Z) \longrightarrow 0; \quad (39)$$

homotopy invariance for vector bundles; Chern classes and their projection formula; and the projective-bundle calculation $\mathrm{CH}^*(\mathbb{P}^{M-1}) = \mathbb{Z}[h]/(h^M)$.

A zero section of a rank- r vector bundle contributes its top Chern class under the vector-bundle identification of Chow groups. Thus, if $E \rightarrow S$ is a rank- r vector bundle over a smooth base,

$$\mathrm{CH}^*(\mathrm{Tot}(E) \setminus S) \simeq \mathrm{CH}^*(S)/(c_r(E)). \quad (40)$$

Indeed, localization and homotopy invariance identify the image of the zero-section pushforward with multiplication by $c_r(E)$. All these operations are integral. In particular, no division by the degree of a finite quotient is permitted.

The following elementary consequence will be central.

Lemma 19.1 (Chow groups of an open affine space). *If W is a nonempty open subset of $\mathbb{A}_k^m$, then $\mathrm{CH}_m(W) = \mathbb{Z}[W]$ and $\mathrm{CH}_i(W) = 0$ for $i < m$.*

Proof. Homotopy invariance gives the assertion for affine space. Restriction to W is surjective by localization, so all lower groups vanish. In top dimension there is one irreducible component, and there are no $(m+1)$ -dimensional subvarieties on which a rational equivalence between top-dimensional cycles could be supported. $\square$

19.2 A finite approximation to the character class

We give an ordinary Chow-group proof of the first ring calculation; one may equivalently phrase it in equivariant Chow theory. This makes the integral coefficient issue and the stabilization range explicit.

Let $E_M = \mathbb{A}_k^M \setminus \{0\}$, with the scalar action $v \mapsto \zeta v$, and let $Q_M = E_M/C_p$. The quotient Q_M is the complement of the zero section in $\mathcal{O}_{\mathbb{P}^{M-1}}(-p)$. To see this, regard E_M as the punctured tautological line bundle $\mathcal{O}_{\mathbb{P}^{M-1}}(-1)^\times$ and take the p th tensor power of a nonzero vector. The fibers of this map are precisely the C_p -orbits. Consequently, (40) gives

$$\mathrm{CH}^*(Q_M) = \mathbb{Z}[h]/(h^M, ph). \quad (41)$$

The character line of $E_M \rightarrow Q_M$ is the pullback of the tautological line or its dual, depending on the convention for the generator. Its first Chern class t_M is therefore h or $-h$. In particular,

$$\mathrm{CH}^*(Q_M) = \mathbb{Z}[t_M]/(pt_M, t_M^M). \quad (42)$$

The choice of sign never changes any ring or nonvanishing assertion below. On each space we consistently define the class by its character line.

19.3 Only the full diagonal removed

Since p is invertible, taking the average of the p points splits the permutation representation:

$$V = F \oplus W, \quad W \simeq \bigoplus_{j=1}^{p-1} (\chi^j)^{\oplus d}, \quad \dim W = N. \quad (43)$$

Here F is the trivial representation of dimension d , and χ is a generating character. Thus $U_0 \simeq \mathbb{A}^d \times (W \setminus \{0\})$.

Proposition 19.2 (Chow ring before deleting partial diagonals). *For $Y = (V \setminus F)/C_p$, with its character class t_Y ,*

$$\mathrm{CH}^*(Y) = \mathbb{Z}[t_Y]/(pt_Y, t_Y^N). \quad (44)$$

Proof. Form the mixed quotient

$$Z_M = (U_0 \times E_M)/C_p.$$

There are two ways to view it. First, it is the complement of the zero section in the rank- M vector bundle $(U_0 \times \mathbb{A}^M)/C_p \rightarrow Y$. The zero section has codimension M , so homotopy invariance and localization give

$$\mathrm{CH}^r(Y) \xrightarrow{\sim} \mathrm{CH}^r(Z_M) \quad \text{whenever } r < M. \quad (45)$$

The character line on Y pulls back to the character line of the common torsor $U_0 \times E_M \rightarrow Z_M$.

Second, project Z_M to Q_M . The decomposition (43) identifies Z_M with a trivial $\mathbb{A}^d$ -factor times the complement of the zero section of the associated rank- N vector bundle $\mathcal{E}_W \rightarrow Q_M$. Its line summands have first Chern classes jt_M , with $1 \leq j \leq p-1$, each repeated d times. A dual convention merely permutes this multiset. Hence

$$c_N(\mathcal{E}_W) = ((p-1)!)^d t_M^N. \quad (46)$$

By (40) and (42),

$$\mathrm{CH}^*(Z_M) = \mathbb{Z}[t_M]/(pt_M, t_M^M, ((p-1)!)^d t_M^N). \quad (47)$$

The coefficient $((p-1)!)^d$ is relatively prime to p . If c denotes this coefficient, choose integers a, b with $ac + bp = 1$. Since $N \geq 1$, $pt_M^N = 0$, and therefore $t_M^N = act_M^N$. Thus the ideal generated by the Euler class is exactly (t_M^N) . Equivalently, Wilson's theorem gives $c \equiv (-1)^d \pmod{p}$.

Choose $M > \max\{N, r\}$. Then (47), together with (45), computes $\mathrm{CH}^r(Y)$ with generator t_Y^r . The identifications preserve products and character classes. Since M can be chosen arbitrarily large, they determine the entire graded ring and give (44). $\square$

The stable-range argument is the standard finite-dimensional classifying-space method of Totaro, also underlying the equivariant construction of Edidin–Graham [14, 15]. What remains is not a formal consequence of the codimension of the full diagonal: the configuration space removes *many additional* diagonals, some of low codimension. We must prove that they do not kill lower powers of t_Y .

20 Deleting partial diagonals creates no new Chow relations

20.1 Prime cyclic symmetry of the partition lattice

A partition π of the label set $\mathbb{F}_p$ specifies the linear subspace

$$V_\pi = \{z_i = z_j \text{ whenever } i, j \text{ lie in one block of } \pi\} \simeq \mathbb{A}_k^{db(\pi)}, \quad (48)$$

where $b(\pi)$ is the number of blocks. Let S_π be the stratum on which π is the *exact* equality partition. Then

$$S_\pi \simeq \text{Conf}_{b(\pi)}(\mathbb{A}_k^d),$$

a nonempty open of V_π . The indiscrete partition corresponds to F ; the discrete partition corresponds to U .

Lemma 20.1 (There are only two invariant partitions). *The only partitions of $\mathbb{F}_p$ fixed by translation by 1 are the discrete and indiscrete partitions. Every other partition has trivial stabilizer under C_p and lies in an orbit of size p .*

Proof. A translation-invariant partition defines a translation-invariant equivalence relation. Let H be the equivalence class of 0. If $a, b \in H$, then $a \sim b$, and translation by $-b$ gives $a - b \sim 0$. Thus H is an additive subgroup of $\mathbb{F}_p$. It is either $\{0\}$ or all of $\mathbb{F}_p$, and the other blocks are its cosets. This gives the two stated partitions. Finally, in a group of prime order, a nontrivial stabilizer would be the entire group. $\square$

For a proper, nontrivial partition, the strata in its orbit are disjoint and form an induced C_p -scheme:

$$\coprod_{g \in C_p} S_{g\pi} \simeq C_p \times S_\pi. \quad (49)$$

Its quotient is therefore isomorphic to S_π itself. In particular, its ordinary Chow groups vanish below its top dimension, by Lemma 19.1. This is where primality performs the decisive work.

20.2 The vanishing pushforward

Let $q_0 : U_0 \rightarrow Y$ be the free quotient map. For a proper nontrivial partition, set

$$W_\pi = V_\pi \setminus F \subset U_0.$$

This is a closed subvariety of U_0 . It is also a nonempty open of the affine space V_π , and has codimension $d(p - b(\pi)) > 0$ in U_0 .

Lemma 20.2 (Transfer-zero for a partial diagonal). *The proper map $W_\pi \rightarrow Y$ has zero pushforward of its fundamental class:*

$$(q_0|_{W_\pi})_*[W_\pi] = 0 \quad \text{in } \text{CH}_*(Y). \quad (50)$$

Proof. Factor the map as the closed immersion $i_\pi : W_\pi \hookrightarrow U_0$ followed by the finite map q_0 . The scheme U_0 is a nonempty open of the affine space V , so its Chow groups below dimension dp vanish. The class $i_{\pi*}[W_\pi]$ lies in the strictly smaller dimension $db(\pi)$ and is therefore zero. Proper pushforward by q_0 preserves this zero. $\square$

No division by p appears. The argument proves zero by factoring a proper pushforward through a zero Chow group, not by multiplying a class by p and then cancelling p . That cancellation would be invalid for the torsion classes we are trying to preserve.

20.3 From the strata to the whole deleted locus

Let

$$C = Y \setminus X = (V \setminus (U \cup F))/C_p.$$

For $1 \leq b \leq p-1$, the locus in V consisting of tuples with at most b distinct points is a finite union of the flats V_π , hence closed. Removing F and taking the finite quotient provides a closed filtration

$$\emptyset = C_1 \subset C_2 \subset \cdots \subset C_{p-1} = C.$$

The difference $C_b \setminus C_{b-1}$ is the disjoint union, over cyclic orbits of partitions with b blocks, of the quotient strata in (49). Each is isomorphic to S_π .

Lemma 20.3 (Generators for the boundary Chow groups). *The groups $\mathrm{CH}_*(C)$ are generated, as abelian groups, by the images of $[W_\pi]$ under $W_\pi \rightarrow C$, where π runs over one representative of every cyclic orbit with $2 \leq b(\pi) \leq p-1$.*

Proof. Apply localization successively to the closed filtration above. On a stratum S_π , the only nonzero Chow group is generated by its fundamental class. The map $W_\pi \rightarrow C$ is proper: it is the restriction of the finite quotient map to a closed subvariety of U_0 . On the dense stratum S_π its map to the quotient stratum is an isomorphism, because the stabilizer of π is trivial. Its fundamental class thus lifts the generator of that quotient stratum with coefficient one. Subtract these lifts from an arbitrary class. Localization leaves a class on the preceding closed piece. Induction finishes the proof.

There can be relations between these generators; no independence claim is needed. The important point is that every generator has a specified proper lift whose pushforward can be evaluated. $\square$

Theorem 20.4 (Partial-diagonal deletion invariance). *Restriction induces an isomorphism of integral Chow rings*

$$\mathrm{CH}^*(Y) \xrightarrow{\sim} \mathrm{CH}^*(X_{p,d}). \quad (51)$$

In particular, (34) holds.

Proof. Localization gives the exact sequence

$$\mathrm{CH}_*(C) \xrightarrow{\iota_*} \mathrm{CH}_*(Y) \longrightarrow \mathrm{CH}_*(X) \longrightarrow 0.$$

By Lemma 20.3, it suffices to evaluate ι_* on the proper lifts of the fundamental classes $[W_\pi]$. Every one of these pushforwards is zero by Lemma 20.2. Thus $\iota_* = 0$, and restriction is an isomorphism. The schemes X and Y are smooth of the same dimension; restriction respects their intersection products. Their character lines also correspond under restriction. Proposition 19.2 therefore gives the claimed ring with precisely the stated generator. $\square$

20.4 What the calculation says in elementary terms

The full diagonal has normal representation consisting of all nontrivial characters, each d times. Its Euler class gives the first relation $t^N = 0$. The remaining equality strata have no cyclic stabilizer, and their classes disappear through ordinary affine-space rational equivalences before they reach the quotient. They therefore contribute no smaller relation.

It follows that

$$\mathrm{CH}^r(X_{p,d}) = \begin{cases} \mathbb{Z}, & r = 0, \\ \mathbb{Z}/p\mathbb{Z}, & 1 \leq r < N, \\ 0, & r \geq N. \end{cases} \quad (52)$$

In particular $\mathrm{CH}^*(X) \otimes \mathbb{Q} = \mathbb{Q}$. A calculation with rational Chow groups would have lost *all* of the positive-degree evidence for the atlas lower bound. When $N > 1$, smoothness gives $\mathrm{Pic}(X) = \mathrm{CH}^1(X) \simeq \mathbb{Z}/p\mathbb{Z}$, recovering the original report's one-dimensional Picard calculation while also determining its higher products.

[write] Second route to Part I's Picard group. Part I's calculation is Theorem 12.3: for odd p and k algebraically closed of characteristic zero, by descent from the explicit unit group (Lemma 12.2). For $d = 1$ and odd p the degree-one part of (52) proves the same group by a different route (localization, the scalar approximation and the Euler class), over any field of characteristic different from p that contains ζ . Both proofs are printed. At $(p, d) = (2, 1)$, $N = 1$ and $\mathrm{CH}^1 = 0$; this case lies outside Part I's hypothesis that p is odd.

21 The exact number of regular Kummer charts

21.1 A cover-nilpotence lemma

The lower bound works for arbitrary covering opens, not merely opens cut out by global sections. We isolate the precise fact needed.

Lemma 21.1 (A trivializing cover kills a Chern power). *Let S be a smooth pure-dimensional variety and let M be a line bundle. If $S = W_1 \cup \cdots \cup W_m$ and $M|_{W_i}$ is trivial for each i , then*

$$c_1(M)^m = 0 \quad \text{in } \mathrm{CH}^m(S). \quad (53)$$

Proof. Let $D_i = S \setminus W_i$ and let $n = \dim S$. We construct a cycle class supported on $D_1 \cap \cdots \cap D_r$ whose pushforward is $c_1(M)^r \cap [S]$. Start with $[S]$ when $r = 0$. Suppose β on $D_1 \cap \cdots \cap D_{r-1}$ has the required pushforward. Apply the first Chern class of the restriction of M to β . On the intersection with W_r , this line bundle is trivial, so the resulting class restricts to zero. Localization lifts it to a class on $D_1 \cap \cdots \cap D_r$. The projection formula shows that its pushforward to S is $c_1(M)^r \cap [S]$.

For $r = m$, the support is empty because the W_i cover S . The resulting class is zero. Identifying homological and cohomological Chow groups on the smooth variety S proves the assertion. $\square$

This proof uses Chern-class operations on the possibly singular closed intersections $D_1 \cap \cdots \cap D_r$; it does not assume that the boundary divisors meet properly, or even that those closed sets are divisors.

Proposition 21.2 (Universal lower bound for the atlas). *Every regular Kummer-chart cover of $X_{p,d}$ has at least N members.*

Proof. By Lemma 18.1, every chart trivializes $\mathcal{L}$, regardless of its chosen nontrivial character. A cover with $m < N$ members would force $t^m = 0$ by Lemma 21.1, contrary to (52). For $N = 1$, a nonempty scheme cannot be covered by zero opens, which is the required lower bound. $\square$

21.2 Fourier charts attain the bound

Write $x_{a,i}$ for the a th coordinate of z_i , where $1 \leq a \leq d$ and $i \in \mathbb{F}_p$. Choose the action on functions so that $\sigma(x_{a,i}) = x_{a,i+1}$, and define

$$u_{a,j} = \sum_{i=0}^{p-1} \zeta^{-ij} x_{a,i} \quad (j \in \mathbb{F}_p). \quad (54)$$

A change of index and the finite geometric-series identity give

$$\sigma(u_{a,j}) = \zeta^j u_{a,j}, \quad x_{a,i} = \frac{1}{p} \sum_{j=0}^{p-1} \zeta^{ij} u_{a,j}. \quad (55)$$

Thus $u_{a,j}$ is a global section of $\mathcal{L}_j$. Let

$$a_{a,j} = u_{a,j}^p \in \Gamma(X, \mathcal{O}_X), \quad D_{a,j} = \{a_{a,j} \neq 0\} \subset X, \quad 1 \leq j \leq p-1.$$

There are exactly $d(p-1) = N$ such opens.

Proposition 21.3 (Optimal Fourier atlas). *The $D_{a,j}$ cover X and are regular Kummer charts, with*

$$(q_* \mathcal{O}_U)|_{D_{a,j}} \simeq \mathcal{O}_{D_{a,j}}[T]/(T^p - a_{a,j}), \quad T \mapsto u_{a,j}. \quad (56)$$

Proof. If all nonconstant modes vanish at a geometric point, Fourier inversion makes each coordinate $x_{a,i}$ independent of i . All p points are then equal, which is impossible in U . Hence the displayed opens cover. On $D_{a,j}$, $u_{a,j}^p$ is a unit. Consequently $u_{a,j}$ is a unit on the inverse image, and it generates the line bundle $\mathcal{L}_j$. Lemma 18.1 gives (56). $\square$

Propositions 21.2 and 21.3 prove $\kappa(X) = N$. Since the lower bound already covers all open sets and all nonlinear character sections, no further argument about special Fourier supports is needed for global optimality.

22 Minimal generators and nonlinear covariant noncompression

22.1 Every nontrivial eigenline needs exactly N sections

Fix $r \in \mathbb{F}_p^\times$. For $j \in \mathbb{F}_p^\times$, let $e_j(r)$ be the unique integer in $\{1, \dots, p-1\}$ with

$$je_j(r) \equiv r \pmod{p}. \quad (57)$$

Then

$$s_{a,j}^{(r)} = u_{a,j}^{e_j(r)} \in \Gamma(X, \mathcal{L}_r), \quad 1 \leq a \leq d, \quad 1 \leq j \leq p-1. \quad (58)$$

Their common zero locus is the same as that of all the nonconstant Fourier modes, hence is empty on X . They give a surjection $\mathcal{O}_X^N \rightarrow \mathcal{L}_r$.

Conversely, m global generators of $\mathcal{L}_r$ produce a cover by their nonvanishing opens, on each of which $\mathcal{L}_r$ and hence $\mathcal{L}$ is trivial. Proposition 21.2 gives $m \geq N$. This proves $\gamma(\mathcal{L}_r) = N$ and completes Theorem 17.1.

There is also a useful direct Chern-class proof of this lower bound. If $\mathcal{O}_X^m \rightarrow \mathcal{L}_r$ has kernel K , then K is a vector bundle of rank $m-1$ and

$$c(K) = \frac{1}{1+rt}.$$

Its degree- m component is $(-rt)^m$ and must vanish. For $m < N$, this contradicts (52), since r^m is prime to p .

Example 22.1 (Four quintic generators of one fixed character). For $p = 5$, $d = 1$, the weight-one eigenmodule is generated by

$$u_1, \quad u_2^3, \quad u_3^2, \quad u_4^4.$$

Indeed, the weights of all four expressions are 1 modulo 5. No three elements of this invertible B -module can generate it, regardless of their polynomial degrees or their allowed denominators in A . Thus the result concerns module generation, not just the inability to omit one entry from this particular list.

22.2 Unavoidable common exceptional points

Theorem 22.2 (Sharp nonlinear character noncompression). *Let $1 \leq m < N$ and let $f_1, \dots, f_m \in \Gamma(U_{p,d}, \mathcal{O}_U)$ be regular functions with*

$$\sigma(f_i) = \zeta^{j_i} f_i, \quad j_i \in \mathbb{F}_p^\times.$$

Then the f_i have a common zero on $U_{p,d}$ after passage to an algebraic closure of k . For $m = N$ there is a family without a common zero, namely the nonconstant Fourier modes.

Proof. Each f_i is a section of $\mathcal{L}_{j_i}$ on X . If they had no common geometric zero, their tuple would be a nowhere-vanishing section of

$$E = \bigoplus_{i=1}^m \mathcal{L}_{j_i}.$$

It would define a trivial line subbundle of E , with a locally free quotient of rank $m - 1$. The top Chern class of E would therefore be zero. But

$$c_m(E) = (j_1 \cdots j_m) t^m \neq 0,$$

by (52). The contradiction proves the first assertion. The Fourier family proves sharpness. $\square$

Equivalently, there is no equivariant regular map from $U_{p,d}$ to the punctured affine representation with fewer than N nontrivial character coordinates. The target coordinate weights may be chosen arbitrarily. In particular, increasing the nonlinear degree of the covariants does not remove this obstruction. In the affine one-dimensional case, regular functions already include expressions with poles along the excluded discriminant; allowing those denominators does not change the result.

This is a more precise statement than saying that familiar radicals sometimes vanish. Any list that is too short has a common exceptional configuration, even when the list is designed by a completely different algebraic construction.

23 Explicit local reconstruction and coherent radical branches

The chart optimum does not by itself explain how roots are recovered. We record the actual local algebra, keeping the denominator and branch conditions visible. For $d = 1$, this is the Fourier–Kummer construction already developed in [12]; its role here is to provide the upper bound and to connect the obstruction with radical solvers.

[write] Relation to Part I. For $d = 1$, Proposition 23.1 below is Theorem 8.1, with (61) equal to (17), and (62) is Part I’s branch transition (18); Part I also gives the coefficient rule (19), which is not restated here. They are printed because they are stated for every d . Index caution: in (59) the first index a' of $c_{a',k}$ is a *coordinate* index and the pivot $u_{b,j}$ is fixed, whereas the first index of Part I’s $c_{j,k}$ is the *pivot*. True reading for $d = 1$: Part II’s $c_{1,k}$ with pivot $u_{1,j}$ is Part I’s $c_{j,k}$. False reading: Part II’s $c_{1,k}$ is Part I’s $c_{1,k}$ (true only for the pivot $j = 1$). In the subsection on changing the pivot, c is at once a coordinate index ($u_{c,h}$) and a ratio. Section 16.2 lists the other overloaded letters.

Choose a pivot $u_{b,j} \neq 0$, and abbreviate

$$u = u_{b,j}, \quad a = u^p, \quad e_j(k) \in \{1, \dots, p-1\}, \quad j e_j(k) \equiv k \pmod{p}.$$

On the chart $D(a)$ define

$$c_{a',k} = \frac{u_{a',k}}{u^{e_j(k)}} \quad (1 \leq a' \leq d, \ 1 \leq k \leq p-1). \quad (59)$$

Every denominator is a unit on this chart. The covariance formula gives

$$\sigma(c_{a',k}) = \zeta^{k-je_j(k)} c_{a',k} = c_{a',k}.$$

Thus all the $c_{a',k}$ and $u_{a',0}$ are regular invariant functions there. Also $c_{b,j} = 1$.

Proposition 23.1 (One shared radical reconstructs the whole tuple). *At a geometric point of $D(a)$, choose any root ρ of $T^p - a$ and put*

$$y_{a',i} = \frac{1}{p} \left(u_{a',0} + \sum_{k=1}^{p-1} \zeta^{ik} c_{a',k} \rho^{e_j(k)} \right). \quad (60)$$

There is an $r \in \mathbb{F}_p$ such that simultaneously for every a' and i ,

$$y_{a',i} = x_{a',i+rj^{-1}}. \quad (61)$$

In particular, all branches produce the same unordered configuration.

Proof. Since $u \neq 0$, all the roots of $T^p - u^p$ are $\rho = \zeta^r u$, with $r \in \mathbb{F}_p$. Substitution gives

$$c_{a',k} \rho^{e_j(k)} = \zeta^{re_j(k)} u_{a',k} = \zeta^{rj^{-1}k} u_{a',k}.$$

Fourier inversion proves (61). The same r works for all coordinates because there was only one radical choice. $\square$

23.1 Changing the pivot

Suppose a second pivot $u_{c,h}$ is also nonzero. On the overlap set

$$e = e_j(h), \quad c = \frac{u_{c,h}}{u_{b,j}^e}, \quad a_{c,h} = c^p a_{b,j}^e.$$

A branch $\rho_{b,j}$ determines the compatible branch

$$\rho_{c,h} = c \rho_{b,j}^e. \quad (62)$$

If $\rho_{b,j} = \zeta^r u_{b,j}$, then $\rho_{c,h} = \zeta^{re} u_{c,h}$. Its reconstructed shift is $reh^{-1} = rj^{-1}$ modulo p , so the ordered reconstructed tuple is unchanged. This supplies exact transition data rather than an independent choice of a new radical on each chart.

23.2 What the local data do not supply

The formula assumes the invariant functions in (59) are already available in the relevant cyclic quotient or intermediate field. It is not an algorithm for computing them from symmetric polynomial coefficients. Indeed, for $d = 1$ the field $\text{Frac}(B)$ is larger than the symmetric coefficient field $k(x_0, \dots, x_{p-1})^{S_p}$. The extension with group C_p appears only after this change of base. Nothing here bypasses the Galois obstruction to solving a generic degree- p polynomial by radicals over the original symmetric coefficient field.

The theorem counts charts on that specified cyclic base. On a single chart only one p th radical is used. Having to cover the entire base with $p - 1$ charts does not mean that $p - 1$ independent p th radicals are needed at any individual input.

24 Ordinary algebra generators: a different exact complexity

The number of Kummer charts is not the number of generators of the finite algebra as an ordinary algebra. We can determine the latter number as well, and the two answers are markedly different.

Definition 24.1. Let $\alpha(q_*\mathcal{O}_U)$ be the least number of global sections $y_1, \dots, y_s$ that generate $q_*\mathcal{O}_U$ as an $\mathcal{O}_X$ -algebra sheaf. Generation is local on the base, but the chosen sections themselves are global. These generators are not required to have character weights.

Theorem 24.2 (No regular equivariant configuration compression). *If $d > s \geq 1$, there is no C_p -equivariant regular morphism*

$$U_{p,d} \longrightarrow U_{p,s}. \quad (63)$$

If $d \leq s$, such a morphism exists by adjoining zero coordinates to each point.

Proof. An equivariant morphism between the free spaces would induce a morphism $f : X_{p,d} \rightarrow X_{p,s}$ pulling back the target torsor, and hence its character line, to the source torsor and character line. Therefore $f^*(t_{p,s}) = t_{p,d}$. Pullback on the Chow rings of the smooth varieties would imply

$$t_{p,d}^{s(p-1)} = f^*(t_{p,s}^{s(p-1)}) = 0.$$

But $0 < s(p-1) < d(p-1)$, so the left side is nonzero by Theorem 17.1. This is a contradiction. The coordinate inclusion when $d \leq s$ visibly preserves distinctness and cyclic permutations. $\square$

Theorem 24.3 (Exact global algebra-generator number). *For the universal prime cyclic configuration cover,*

$$\alpha(q_*\mathcal{O}_U) = d. \quad (64)$$

Proof. For the upper bound, take $x_{1,0}, \dots, x_{d,0}$, the coordinates of the point with label 0. On a geometric fiber of q , their simultaneous values are precisely the p distinct points $z_0, \dots, z_{p-1}$. Polynomial functions in these d coordinates separate those points and generate the full product algebra of functions on the fiber, by the Chinese remainder theorem for their distinct maximal ideals.

To pass from fibers to sheaf generation, note that each multiplication operator by $x_{a,0}$ has a monic characteristic polynomial of degree p . By Cayley–Hamilton, the monomials $\prod_a x_{a,0}^{e_a}$ with $0 \leq e_a < p$ span the subalgebra generated by the chosen sections. They define a morphism from a finite free sheaf to $q_*\mathcal{O}_U$ that is surjective on every geometric fiber. Its coherent cokernel is zero by Nakayama’s lemma. Thus d generators suffice.

For the lower bound, suppose s global functions $y_1, \dots, y_s$ on U generate the algebra. On each geometric fiber their simultaneous values must distinguish all p points: otherwise every polynomial in them would take equal values at two of the factors of the product algebra. The functions

$$(\sigma^i y_1, \dots, \sigma^i y_s), \quad i \in \mathbb{F}_p,$$

therefore define an equivariant regular morphism $U_{p,d} \rightarrow U_{p,s}$. Theorem 24.2 forces $s \geq d$. Zero generators cannot generate a rank- $p > 1$ algebra, so this also covers the remaining possibility $s = 0$. $\square$

24.1 The one-dimensional cover is globally monogenic

For $d = 1$, define the invariant polynomial

$$f(T) = \prod_{i=0}^{p-1} (T - x_i) \in B[T]. \quad (65)$$

There is an isomorphism

$$B[T]/(f(T)) \xrightarrow{\sim} A, \quad T \mapsto x_0. \quad (66)$$

For a direct proof, base change along the faithfully flat map $B \rightarrow A$. All pairwise differences $x_i - x_j$ are units in A , so the Chinese remainder theorem identifies the source after base change with A^p . The torsor identity identifies $A \otimes_B A$ with the same product, and the map evaluates T at the p cyclic translates of x_0 . It is an isomorphism after this base change, hence before it.

Thus the original cover has one global *ordinary* polynomial coordinate while requiring $p - 1$ regular *Kummer* charts. The obstruction is to the special equation $T^p - a$ with an eigen-coordinate, not to every possible global primitive element.

24.2 The regular-representation bundle detects the difference

There is also an informative Chern calculation. Let $E = q_*\mathcal{O}_U$. Since $E = \bigoplus_{j=0}^{p-1} \mathcal{L}_j$, the identity $\prod_{j=1}^{p-1} (1 + jz) = 1 - z^{p-1}$ over $\mathbb{F}_p$ gives, integrally in our Chow ring,

$$c(E) = 1 - t^{p-1}. \quad (67)$$

The intermediate coefficients vanish because they are divisible by p and every positive power of t is p -torsion. For $d \geq 2$, $t^{p-1} \neq 0$, so E is not a free vector bundle. In particular, it cannot have a global monic one-variable polynomial presentation. For $d = 1$, that Chern class vanishes, consistently with the explicit free basis $1, x_0, \dots, x_0^{p-1}$ in (66). This Chern observation alone proves non-monogenicity for $d \geq 2$; Theorem 24.3 supplies the exact stronger value d .

25 Allowing partial collisions does not improve the atlas

The deletion theorem has a useful consequence that goes beyond the pairwise-distinct configuration space.

Theorem 25.1 (Intermediate-open invariance). *Let $W \subset V_{p,d}$ be any C_p -invariant open satisfying*

$$U_{p,d} \subseteq W \subseteq V_{p,d} \setminus F. \quad (68)$$

Then, for $Z = W/C_p$ and its generating character line $\mathcal{L}_Z$,

$$\mathrm{CH}^*(Z) = \mathbb{Z}[t_Z]/(pt_Z, t_Z^N), \quad \kappa(Z) = \gamma((\mathcal{L}_Z)_j) = N \quad (j \neq 0). \quad (69)$$

Proof. There are open immersions $X \subseteq Z \subseteq Y$. Both restriction maps in

$$\mathrm{CH}^*(Y) \longrightarrow \mathrm{CH}^*(Z) \longrightarrow \mathrm{CH}^*(X)$$

are surjective by localization. Their composite is an isomorphism by Theorem 20.4. The first map must therefore be injective as well, and then the second is an isomorphism. Character classes restrict compatibly, giving the ring formula.

The cover-nilpotence argument gives the lower bound N for Kummer charts. The same N Fourier modes have no common zero anywhere outside F , so they supply the upper bound on Z . Their powers of a fixed nonzero weight similarly give the exact eigenline generator number. $\square$

Corollary 25.2 (Cyclic graph configurations). *Let Γ be any nonempty simple graph on $\mathbb{F}_p$ invariant under translation. Impose only the inequalities $z_i \neq z_j$ for edges $\{i, j\}$ of Γ . The resulting free cyclic quotient has the same Chow ring and the same optimal Kummer-chart number $d(p - 1)$.*

Proof. The open in question contains the pairwise-distinct configuration and excludes the full diagonal, because the graph has an edge. Apply Theorem 25.1. $\square$

For example, imposing only consecutive cyclic inequalities $z_i \neq z_{i+1}$ already has the same obstruction as imposing every pairwise inequality. Thus adding a large family of allowed partial collisions does not create a smaller global Kummer atlas.

The direction of the hypothesis is essential. If one deletes additional points from U , for instance by keeping only $D(u_{a,j}^p)$, a single Kummer chart does work. Our theorem concerns intermediate opens *containing all distinct configurations*; it does not assert invariance under arbitrary further deletions. Likewise, the ordinary algebra-generator conclusion (64) is not asserted for these larger opens, since the point-coordinate values may now repeat.

26 Explicit torsion cycles and small-degree examples

26.1 Geometric representatives of every nonzero power

The obstruction is not merely a formal symbol in a presentation. Its powers admit very concrete cycle representatives.

Proposition 26.1 (Fourier-coordinate cycles). *Let S be a subset of the N index pairs (a, j) with $j \neq 0$, and write $r = |S| < N$. The equations $u_{a,j} = 0$ for $(a, j) \in S$ define a nonempty smooth integral closed subvariety Z_S of $X_{p,d}$ of codimension r . Its Chow class is*

$$[Z_S] = \left(\prod_{(a,j) \in S} j \right) t^r. \quad (70)$$

For $1 \leq r < N$ it is a generator of $\mathrm{CH}^r(X) \simeq \mathbb{Z}/p\mathbb{Z}$.

Proof. The Fourier transform is an invertible linear change of coordinates on V . Thus these equations define a linear subspace of codimension r . Its intersection with U is a smooth integral open subset. It is nonempty: retain any one of the nonconstant modes not in S , set it to a nonzero value, and set the other nonconstant modes to zero. Fourier inversion gives p distinct points, since a nonzero character of C_p takes p distinct values.

The intersection descends to a smooth integral closed subvariety of X , and regular-embedding codimension can be checked after the étale cover $U \rightarrow X$. Its equations are the components of a regular section of $\bigoplus_{(a,j) \in S} \mathcal{L}_j$. Its fundamental class is the top Chern class of this bundle, which is (70). The coefficient is nonzero modulo p , proving the last assertion. $\square$

26.2 A pure-mode locus represents the sharp obstruction

Take S to contain every nonconstant index except one, (b, h) . The resulting locus has all nonconstant modes zero except $u_{b,h} \neq 0$. Before quotienting it is $\mathbb{A}^d \times \mathbb{G}_m$, and the cyclic group scales the second factor with a nontrivial weight. Its quotient is again $\mathbb{A}^d \times \mathbb{G}_m$, with coordinate $u_{b,h}^p$ on the multiplicative factor. For $N > 1$, this very simple subvariety represents a nonzero multiple of t^{N-1} , of exact order p in the ambient Chow group.

When $d = 1$, its root configurations are

$$x_i = \eta + \beta \zeta^{hi}, \quad \beta \neq 0,$$

with root polynomial

$$\prod_i (T - x_i) = (T - \eta)^p - \beta^p. \quad (71)$$

Thus the highest nonzero obstruction power is represented by a locus of translated binomial configurations. The cover itself is easy to write on that locus; its nonzero *ambient cycle class* is what obstructs covering the whole space with too few charts. There is no contradiction between these two statements.

26.3 Numerical values and the exceptional smallest case

p	d	N	Integral Chow ring	Kummer charts	Algebra generators
2	1	1	$\mathbb{Z}$	1	1
3	1	2	$\mathbb{Z}[t]/(3t, t^2)$	2	1
5	1	4	$\mathbb{Z}[t]/(5t, t^4)$	4	1
7	1	6	$\mathbb{Z}[t]/(7t, t^6)$	6	1
3	2	4	$\mathbb{Z}[t]/(3t, t^4)$	4	2
5	2	8	$\mathbb{Z}[t]/(5t, t^8)$	8	2

The last column concerns the distinct-point cover only, as in Theorem 24.3. The chart count equals the eigenline generator count, not the ordinary algebra-generator count.

For $p = 2, d = 1$, the single nonconstant Fourier mode is $x_0 - x_1$, which is already a unit on the entire configuration space. It gives the global Kummer coordinate predicted by $N = 1$.

For a quintic, $t^3 \neq 0$ but $t^4 = 0$. The product coefficient for the pure weight-one locus is $2 \cdot 3 \cdot 4 = 24 \equiv -1 \pmod{5}$, so that locus represents $-t^3$. Three proposed nonlinear charts cannot cover $X_{5,1}$, because their trivializations would force this class to vanish. For a septic, the analogous obstruction is $t^5 \neq 0$; the exact optimum is six, not merely a bound between two and six.

27 Proof audit, reproducible checks, and formalization

27.1 The logical dependency boundary

The main result is not inferred from finite experiments. Its proof has the following dependency structure.

Conclusion	Essential input	Evidence supplied
Punctured-quotient Chow ring	Vector-bundle localization, scalar C_p approximation, Euler class	Written proof, Proposition 19.2
Partial-diagonal deletion invariance	Prime cyclic partition lemma, ordinary Chow groups of open affine spaces, proper pushforward	Written proof, Theorem 20.4
Optimal arbitrary-open atlas	Nonzero powers of $c_1(\mathcal{L})$, cover-nilpotence, Fourier cover	Written proof, Section 21
Eigenline and algebra generators	Explicit sections, torsor fibers, configuration noncompression	Written proofs, Sections 22 and 24
Finite orbit and Fourier identities	Integer and finite-field arithmetic	Executed Python checks; not kernel verification

The classical Chow-theoretic properties are mathematical inputs, not newly formalized infrastructure. The finite checker supplies no evidence for localization, Chern-class functoriality, or the nonvanishing of a Chow class. Those assertions are established by the written arguments using standard intersection theory [16, 14, 15].

Several tempting but invalid shortcuts have been avoided. We do not cancel p in a torsion group. We do not infer the nonlinear optimum from coordinate-cover irredundancy. We do not

identify an ordinary monogenic presentation with a Kummer presentation. We do not confuse trivializing an eigenline with splitting the torsor. We do not apply the prime partition lemma in composite degree.

27.2 Exact checks included in the package

The finite checker, `code/verify_finite_checks.py`, uses only Python’s standard library. It writes `data/finite_checks.json`.³ The recorded run used Python 3.13.5 and completed successfully. It enumerates all set partitions for $p = 2, 3, 5, 7$, checks the cyclic orbits and the two fixed partitions, and compares the counts with independently computed Stirling numbers. For the proper partitions, the numbers of cyclic orbits are:

p	Total partitions	Proper-partition orbit counts by increasing block number
2	2	none
3	5	1
5	52	3, 5, 2
7	877	9, 43, 50, 20, 3

The program also checks that the proper partition $\{\{0, 2\}, \{1, 3\}\}$ is fixed by the cyclic action in degree four, a concrete warning against dropping primality.

The Fourier tests are exact finite-field calculations, not floating-point approximations. They check inversion, character covariance, the common zero locus of the nonconstant modes, the fixed-character powers (58), invariant pivot ratios, and all branches on a selected group of inputs.

p	Field	Input coverage	Tuple checks	Branch checks
3	$\mathbb{F}_7$	exhaustive	343	384
5	$\mathbb{F}_{11}$	exhaustive	161051	640
7	$\mathbb{F}_{29}$	seeded sample	4096	896

For each row, branch checks use all p branches on 128 nonconstant inputs; they are not exhaustive branch checks on all tuples. Additional tests examine 256 vector-valued tuples for each of the six pairs (p, d) with $p \in \{3, 5, 7\}$ and $d \in \{2, 3\}$, as well as Euler coefficients for six primes and four dimensions. The report records precisely which tests were exhaustive and which were sampled. The sampling seed is fixed for reproducibility.

The general partition lemma and the Fourier identities have already been proved in the text. These tests are meant to catch normalization, index, and implementation errors, especially the sign of the Fourier transform and the branch-shift exponent. They do not replace those proofs and are not described as formal certificates for the article’s geometric conclusions.

27.3 A realistic formalization decomposition

A proof-assistant implementation should separate the elementary and intersection-theoretic layers. The following are proposed milestones, not declarations claimed to exist in the repository.

The first layer is finite and algebraic: define the prime cyclic action on equality partitions; prove Lemma 20.1; verify the Fourier character projectors, inversion, and the pivot branch shift; and prove that all nonconstant modes vanish exactly on the full diagonal. These statements need finite group actions, field algebra, and finite sums, but no Chow theory.

³[write] Shipped as `code/02-kummer-atlases-verify_finite_checks.py` and `data/02-kummer-atlases-finite_checks.json`. Its default output path is `data/finite_checks.json` beside the script’s parent directory; always pass `--output` as in the note at the end of Appendix C.

The next layer is geometric: construct the free quotient, its character line bundles, and the multiplication isomorphisms; prove the Kummer chart equivalence; and formalize the implication from a finite trivializing cover to vanishing of a Chern power. The last implication can be isolated as a generic theorem, independent of configuration spaces.

The final layer is intersection-theoretic: establish the scalar finite approximation, its stable range, the Euler-class quotient, and the zero-pushforward boundary argument. The ordinary Chow proof in Section 20 is deliberately arranged to avoid building an equivariant intersection theory for every singular partial-diagonal union. A sufficient verified interface would consist of localization, proper pushforward, homotopy invariance, and first/top Chern-class operations with their required compatibilities.

No audit of the current coverage of these facilities in a particular proof-assistant library was performed for this manuscript. A file that postulates them would be a conditional formalization, not a completed kernel proof. The delivered package contains no Lean or Rocq code.

28 Further research questions and topics

The original nonlinear-cover question is settled in the model specified here. The questions below concern genuine extensions or refinements of that model. They are proposed research targets, not assertions that each has already been posed as an open problem in the published literature.

Research question 28.1 (Composite-degree cyclic configurations). For composite n , determine the integral Chow ring of $\text{Conf}_n(\mathbb{A}^d)/C_n$ and the optimal regular Kummer-chart count when $\text{char}(k) \nmid n$ and $\mu_n \subset k$.

The action on distinct configurations remains free, but proper equality partitions can have nontrivial stabilizers. Their quotient strata need not have trivial positive-codimension Chow groups, so the deletion proof changes essentially. The degree-four partition in the finite checks is the first obstruction. An actionable first case is $n = 4$, with separate tracking of the order-two and primitive order-four character pieces. The prime-degree formula should not be extrapolated without this work.

Research question 28.2 (Descent without a primitive root of unity). What is the best replacement for an optimal character-line atlas over a field that does not contain μ_p ? How much of the atlas and its transition data descends from the cyclotomic extension?

The constant group C_p and the group scheme μ_p must be separated. A local binomial presentation is not automatically an available model for every constant cyclic torsor over a nonsplitting base. A precise formulation should use Galois descent of the character pieces or their higher-rank sums, rather than silently reuse Lemma 18.1. The split-base result supplies a lower-bound test after scalar extension but not a complete descended construction.

Research question 28.3 (Other permutation groups and solvable towers). For a subgroup $H \leq S_n$, which products of character Chern classes survive on $\text{Conf}_n(\mathbb{A}^d)/H$, and what exact local radical-coordinate complexity do they force along a solvable subgroup series?

The partition filtration still exists, but strata now contribute through their stabilizers. The natural next targets are prime-degree affine groups and small solvable imprimitive groups. This problem goes beyond computing Pic: higher products are needed for sharp cover counts. A useful deliverable would identify the full image of each stabilizer's cycle pushforward, with integral coefficients throughout.

Research question 28.4 (Coefficient-efficient optimal atlases). Can the optimal $p - 1$ charts for $d = 1$ be presented with substantially smaller exact invariant formulas, while preserving all specialization and branch conditions?

The chart count cannot improve, but expression degree, bit complexity, intermediate-field representation, and certificate size can. The quintic and septic pivot ratios are concrete test cases. Any proposed algorithm should compute its invariant data from the intended input field rather than assume an ordered root tuple. The geometric lower bound does not preclude significant algorithmic improvement here.

Research question 28.5 (Regular versus holomorphic or continuous chart covers). For $k = \mathbb{C}$, compare the optimal regular Zariski Kummer atlas with holomorphic and continuous trivializing covers of the associated character line bundle.

The present proof uses algebraic Chow classes. To transfer it to another category, one must understand their cycle-class images and the relevant notion of an open cover. Equality of the algebraic and topological numbers is not a consequence of this paper. Computing the image of t^r under realization, beginning with $p = 5, d = 1$, would clarify which part of the obstruction is specifically algebraic.

Research question 28.6 (Beyond Chow groups: what changes under partial collisions?). The inclusions in (68) induce isomorphisms of Chow rings. How do their higher Chow groups, algebraic cobordism groups, or motives differ?

The removed strata are nonempty, so an isomorphism of ordinary Chow rings should not be mistaken for an equivalence of all invariants. The proof identifies explicit boundary strata and zero ordinary pushforwards, providing a concrete starting point for studying higher localization maps. A first test is the comparison between consecutive-edge and complete-graph configurations in degree five.

Research question 28.7 (Explicit rational equivalences for the obstruction cycles). Can the relations $p[Z_S] = 0$ and the transfer-zero identities be realized by small, explicit rational-equivalence chains suitable for a verified cycle checker?

The current proof derives these relations through general Chow operations. Making the chains explicit could reduce the trusted geometric interface needed for the first concrete case. The pure-mode surface representing t^3 for quintics is a particularly small target. Such a chain must also be paired with a genuine nonvanishing argument; exhibiting p -torsion alone does not prove exact order p .

Research question 28.8 (Global generators in other configuration families). For graph configurations or other free invariant opens, determine the minimum number of global ordinary algebra generators of the finite cover, and compare it with the unchanged Kummer-chart count.

On the distinct-point space that number is exactly d . When partial collisions are allowed, the coordinate functions at one label can fail to distinguish an entire cyclic fiber. The Chow ring remains the same under (68), but that fact alone supplies no matching upper bound for ordinary algebra generation. This separates a new constructive problem from the atlas problem already solved here.

29 Conclusion

The sharp answer to the repository's nonlinear-cover problem is $p - 1$. Its explanation is the entire sequence of nonzero powers

$$t, t^2, \dots, t^{p-2} \quad \text{in} \quad \text{CH}^*(\text{Conf}_p(\mathbb{A}^1)/C_p),$$

not merely the first Picard-group obstruction. The higher-dimensional formula is $d(p-1)$, and the proof remains valid for every split base field of characteristic different from p .

The essential geometric step is that all proper partial-diagonal strata have trivial cyclic stabilizer and push forward to zero through the ordinary Chow groups of an open affine space. Consequently, removing those strata preserves the characteristic classes of the punctured regular representation. A general cover-nilpotence lemma converts those classes into a sharp lower bound, matched by explicit Fourier charts.

The same calculation controls nonlinear covariants, exhibits concrete torsion cycles, and determines the exact ordinary algebra-generator number d . In particular, a one-dimensional cyclic cover can have one global ordinary polynomial coordinate while needing $p-1$ regular Kummer charts. Keeping these different notions of complexity separate is essential for interpreting what an optimal radical atlas actually proves.

C Provenance, literature comparison, and reproducibility

C.1 Repository snapshot

The relevant repository text was checked at the full commit

1ee53d57de253d16cdaad79d1f54bbd95d76d682.

The selected source is

SetTheory/Cardinals/docs/reports/galois-theory-and-radicals/specialization
-safe-radical-solvers/article.tex.

Its inspected Git blob is f8303298d0ed04eace36343ec5208c9dbe127042. The question answered here is Section 14, Research Question 14.5 of that report. Its existing results distinguish the nonexistence of one global Kummer coordinate, its Picard-group calculation, and minimality within the coordinate Fourier cover. Our arbitrary-open lower bound addresses precisely the gap left by those results.

The repository was changing during inspection; the commit above is an immutable anchor rather than a claim that a moving default branch will retain the same contents. The adjacent report directory was inspected, and targeted repository searches for the Kummer/Chow connection were performed. This was not a mathematical audit of every research archive or of all concurrent work in the repository.

C.2 Relationship to the external literature

Totaro's construction of $CH^*(BG)$ and his analysis of cyclic products supply close methodological precedents [14]. The scalar approximation and Euler-class quotient in Proposition 19.2 belong to this established framework. Edidin–Graham provide the systematic equivariant intersection theory behind the same calculation [15]; ordinary intersection theory supplies the localization and Chern-class operations [16]. The eigenline interpretation of a Kummer torsor is also standard [3].

The explicit contribution developed here is the combination of the partial-diagonal zero-pushforward proof with the sharp arbitrary-open atlas lower bound and its concrete consequences for the specified repository question. The manuscript deliberately supplies a proof rather than infer novelty from a search failure. A specialist literature review could identify prior formulations or stronger existing results; such a discovery would change the attribution, not the logical role of the proof as an answer to the repository question.

C.3 Package contents and rebuild

The package contains the standalone LaTeX source and compiled PDF, this manuscript's README, the finite checker, its JSON output, and a source manifest. The bibliography is embedded in the `.tex` file; no BibTeX database or external graphics are needed.

Run the finite checks with

```
python3 code/verify_finite_checks.py
```

and compile with

```
pdflatex -interaction=nonstopmode -halt-on-error optimal_kummer_atlases.tex
pdflatex -interaction=nonstopmode -halt-on-error optimal_kummer_atlases.tex
```

A third pass may be needed after changes to the table of contents. The included `build.sh` uses `latexmk` when available and otherwise runs three PDFLaTeX passes. No network access is needed to run the checker or build the article once the TeX packages are installed.

[write] Shipped names and a safe rerun. The paragraph above describes the delivered package. In this report the checker is `code/02-kummer-atlases-verify_finite_checks.py`, its recorded output `data/02-kummer-atlases-finite_checks.json`, the source manifest `data/02-kummer-atlases-source_manifest.json`, the build record `data/02-kummer-atlases-build_check.json` (which describes the delivered 22-page PDF, not this report's PDF), and the build script `code/02-kummer-atlases-build.sh`. The manuscript's source, PDF and README are not shipped; this report is built from `article.tex`. The build script does not work in the shipped layout: it changes to its own directory `code/` and then calls `code/verify_finite_checks.py` and compiles `optimal_kummer_atlases.tex`, neither of which exists there. In the delivered layout it reran the checker in place, overwriting the recorded `data/finite_checks.json`. Run without `--output`, the checker writes `data/finite_checks.json` relative to the report directory, an unshipped file beside the shipped record; on Windows its `write_text` emits CRLF line endings. Run from the report directory, writing to a scratch directory:

```
py code/02-kummer-atlases-verify_finite_checks.py --output <tmpdir>/finite_checks.json
```

At the write this exited with `PASS` in about 9 s, and after removing carriage returns the fresh file equalled `data/02-kummer-atlases-finite_checks.json` except for the field `"python_version"` (3.14.4 at the write, 3.13.5 recorded).

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